National Institute Of Business Management

Diploma In Software Engineering

GADSE251F Batch

Web-Based Tourism Services Marketplace Platform

Final Project Report

("Ceylon Tour Mate.lk")

Submitted by:
Name: K.K.Naveen Rashmika Devinda
Registration No: GADSE251-F-018

Submission Date: 2026-07-27

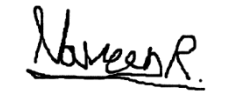
National Institute of Business Management (NIBM) – Galle Campus

No. 132, Samudhra Mawatha (Sea Street), Galle 80000, Sri Lanka

Declaration

I, Naveen Rashmika Devinda, hereby declare that this final project report submitted for DSE final project is an original work done by me under the guidance of Ms.Ashanthi Kurakulasuriya ,also declare that this report record has not been submitted to any other institute.

Signature:



.....

Naveen Rashmika Devinda

GADSE251F-018

Date:

2026.07.27

Supervised by:

Ms. Ashanthi Kurakulasuriya

(Consultant/Lecturer - National Institute of Business Management)

Contents

Chapter 1: Introduction 6
1.1 Introduction of the Organization 6
1.2 Organization Structure 6
1.3 Current Operations in Organization 7
1.4 Users and Responsibilities in Organization 7
1.5 Problem Definition 7
1.6 Project Objectives 8
1.7 Proposed Solution 8
1.8 Chapter Summery 8
Chapter 2: Methodology 9
2.1 Introduction 9
2.2 Data Collection Methods 9
2.3 Software Process Model 9
2.4 Software Development Tools 10
2.5 Testing Strategies 10
2.6 Implementation Plan 10
2.7 Chapter Summary 11
Chapter 3: Analysis 11
3.1 Introduction 11
3.2 UML Diagrams 11
3.2.1 Use Case Diagram of Current System 11
3.2.2 Use case Diagram of Proposed System 12
3.2.3 Class Diagram of Proposed System 13
3.2.4 Sequence Diagrams of Proposed System 14
3.3 ER Diagram of Proposed System 17
3.4 Chapter Summery 17
Chapter 4: Solution Design 18
4.1 Introduction 18

4.2 Interface Design 18
4.3 Database Design 29
4.4 Report Layout Design 34
Chapter 5: Implementation 37
5.1 Introduction 37
5.2 System architecture 37
5.3 Main Coding Areas 38
5.4 Admin Report Implementation 39
Chapter 6: Testing 40
Chapter 7: Conclusion 41
7.1 Achievements 41
7.2 Limitations 41
7.3 Future Developments 41
7.4 Final Conclusion 41
References 42

Figure 1 Organization Structure..... 6

Figure 2 Use Case Diagram of Proposed System..... 12
Figure 3 Class Diagram of Proposed System..... 13
Figure 4 Sequence Diagram for Admin Approve/Reject Listings..... 14
Figure 5 Sequence Diagram for Admin Approve/Reject Registrations..... 15
Figure 6 Sequence Diagram for Tourist Booking a Listing..... 16
Figure 7 ER Diagram of Proposed System..... 17
Figure 8 Interface 1..... 18
Figure 9 Interface 3..... 19
Figure 10 Interface 2..... 19
Figure 11 Interface 4..... 20
Figure 12 Interface 5..... 20
Figure 13 Interface 6..... 21
Figure 14 Interface 7..... 21
Figure 15 Interface 9..... 22
Figure 16 Interface 8..... 22
Figure 17 Interface 10..... 23
Figure 18 Interface 11..... 23
Figure 19 Interface 12..... 24
Figure 20 Interface 13..... 24
Figure 21 Interface 14..... 25
Figure 22 Interface 15..... 25
Figure 23 Interface 15..... 26
Figure 24 Interface 16..... 26
Figure 25 Interface 18..... 27
Figure 26 Interface 17..... 27
Figure 27 Interface 19..... 28
Figure 28 Interface 20..... 28
Figure 29 report layout 1..... 34
Figure 30 report layout 2..... 35
Figure 31 report layout 3..... 35
Figure 32 report layout 4..... 36
Figure 33 report layout 5..... 36
Figure 34 Main classes..... 37
Figure 35 Welcome message..... 38
Figure 36 Session class..... 38
Figure 37 mPDF..... 39

Chapter 1: Introduction

1.1 Introduction of the Organization

Tourism has become one of the fastest growing industries in Sri Lanka. Every year it keeps growing, according to the Sri Lankan Tourism Development Authority about 2.3 million tourists visited Sri Lanka in last year exceeding previous records.

With this growth a lot of opportunities has created for Sri Lankans including hotel owners, tour guides, shop owners and local experience providers. They offer tourism related services as their main income.

For tourists getting these information become bit hard and tricky due to spread of these info on many platforms.

So I thought of making a centralized platform named "**CeylonTourMate.lk**" to connect tourists and service providers across Sri Lanka.

1.2 Organization Structure

These are the main roles in the platform

- Admin
- Tourist
- Tour Guide
- Hotel Owner
- Shop Owner
- Experience Provider

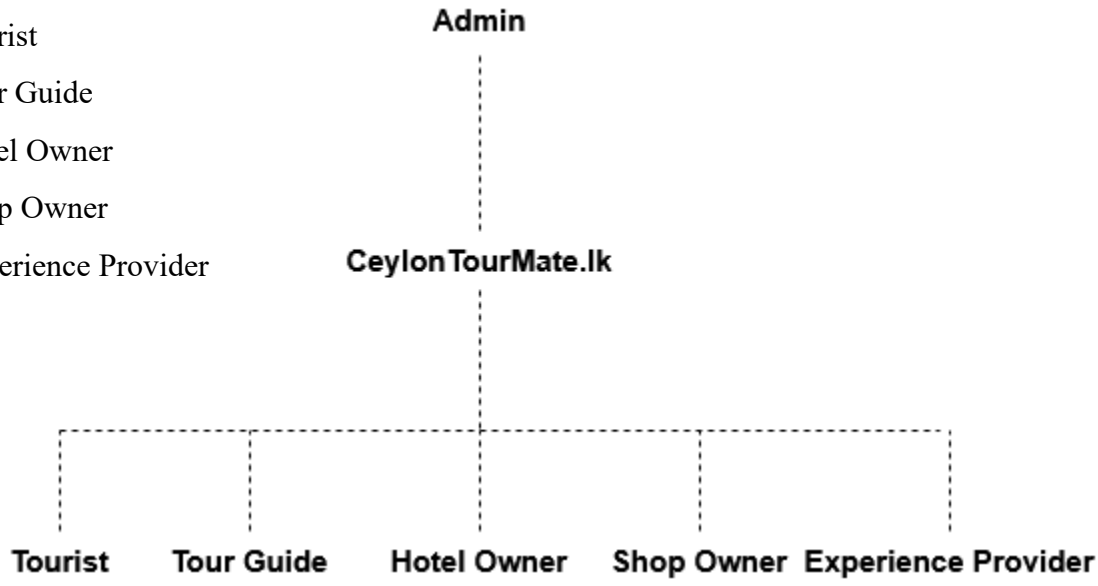


Figure 1 Organization Structure

1.3 Current Operations in Organization

Currently information about tour guides, hotels, traditional shops and local experiences in Sri Lanka are spread across multiple platforms like social media, third party booking sites.

Also there isn't any centralized platform for tourists to get these information from verified local service providers as well as local businesses to list their listings for international audience.

1.4 Users and Responsibilities in Organization

Role	Responsibilities
Admin	Approve/reject partner registrations and listings, manage users, set commission and listing fee rates, monitor bookings, generate reports
Tourist	Search/browse listings, make bookings, send messages to partners, write reviews, complete payments
Tour Guide	Manage listings, set availability, handle bookings, communicate with tourists, view earnings
Hotel Owner	Manage listings, manage rooms availability, handle bookings, communicate with tourists, view earnings
Shop Owner	Manage listings, set operating hours, pay monthly listing fees, reply to messages
Experience Provider	Manage listings, set availability, handle bookings, communicate with tourists, view earnings

1.5 Problem Definition

At the moment tourists who visit Sri Lanka face several challenges.

- **Overpriced and unlicensed guides**- Tourists specially first time visitors sometimes meet overpriced or unlicensed guides who try to overcharge or scam them, which is going to be a bad experience for them.

- **Hidden places**- Sri Lanka has many small villages and attraction places that are not included on current tourism platforms, leaving tourist not knowing information about them.
- **Language barrier**- this is the most common issue when it comes to traveling. Not every Sri Lankan can speak English as well as every tourist can speak Sinhala and Tamil. This can be bit challenging when getting a service.
- **Finding services**- Common services like transportation, places to stay and experiences are now spread over many platforms and it can be difficult for both tourists and people who are willing to provide these services.

1.6 Project Objectives

My project objectives are,

- To develop and design a centralized platform to connect tourists with verified local Hotel/Villa owners, Tour guides, Shop owners and Experience providers in Sri Lanka.
- To give small local business a chance to list their listing to the international audience.
- To minimize the risk of tourists getting scammed and making it easy to get those services through one trusted platform.
- To earn from the platform by getting a commission from bookings and listing fee from shop listings.

1.7 Proposed Solution

CeylonTourMate.lk is a centralized platform that built for the Sri Lankas tourism industry that provides all kinds of trustworthy information about local services to tourists.

1.8 Chapter Summery

In this chapter I introduced my project, CeylonTourMate.lk along with the problem of current industry. With the growing of tourism industry, for problems faced by tourists and locals, and as a solution for that I proposed a centralized platform to connect them all.

Chapter 2: Methodology

2.1 Introduction

In this chapter im going to describe how I gather information for my project and approaches I used to develop it. This covers data collection, software development model, tools I used, technologies and testing methods.

2.2 Data Collection Methods

Since my project is a self created project rather than a sponsored one, I had to do following methods to collect data.

- Current market analysis – I did reviews on existing tourism platforms, gathered information from Sri Lankan Tourism Development Authority, viewed attractional places on google maps, and read articles to understand the current gap in the industry.
- Personal knowledge and observation – I found current tourists problems on social media like unlicensed guides, language barrier , etc.. and had a common knowledge of other challenges to tourists.
- Supervisor consultations – since the project started I had meetings with my supervisor Ms.Ashanthi Kurakulasuriya got feedbacks and recommendations throughout development of the project.

2.3 Software Process Model

To develop CeylonTourMate.lk I used the **Waterfall Method,** which is a clear, structured way to develop, since I had a clear idea of my project and its requirements.

The stages of the waterfall method are,

1. **Requirements Analysis** – in this stage I gathered all the requirements for the system and made the SRS document.
2. **System Design** – in this stage I designed all the diagrams.
3. **Coding** – in this stage I started to coding the platform.
4. **Testing** – in this stage I tested my system.
5. **Deployment** – in this stage I hosted my web on a live server.

2.4 Software Development Tools

Category	Tool
Frontend	Notepad++(HTML,CSS,JS)
Backend	Notepad++(PHP)
Database	MySQL(PhpMyAdmin)
Local Server	XAMPP
Code Editor	Notepad++
Version Control	GitHub
Documentation	Microsoft Word
Diagramming	Draw.io
Report Generation	mPDF

2.5 Testing Strategies

I applied following testing strategies to my project.

- Unit Testing – using this I tested individual php files to confirm their work.
- Functional Testing – tested each functions in the system to confirm their work.
- Validation Testing – tested all form inputs and validations.
- Regression Testing – after found some bugs I fixed them and retested them.

2.6 Implementation Plan

My implementation plan was as following order.

1. Gathering all the requirements and making the SRS document
2. Use case diagram drawing
3. ER diagram designing
4. Class diagram designing
5. Made the database in MYSQL as the ER diagram
6. Coded all the classes like in class diagram
7. Made the Main interface home.html
8. Admin panel coding with all other interfaces

9. Tourist panel coding with all other interfaces
10. Tour guide panel coding with all other interfaces
11. Shop Owner panel coding with all other interfaces
12. Hotel Owner panel coding with all other interfaces
13. Experience Provide panel coding with all other interfaces
14. Testing bug fixing
15. Final report documentation and presentation making

2.7 Chapter Summary

In this chapter I described how I gathered my data to develop my project, software development model, tools I used for development and documentation, testing strategies, and overall methodology of my project.

Chapter 3: Analysis

3.1 Introduction

In this chapter im going to present my analysis I did for CeylonTourMate.lk, including usecase diagram, class diagram, sequence diagram and ER diagram. These diagrams show how the platform works and its database works.

3.2 UML Diagrams

3.2.1 Use Case Diagram of Current System

Since I made this project to fill a gap in the industry, where services are currently spread across multiple platforms, there is not any current system.

3.2.2 Use case Diagram of Proposed System



Figure 2 Use Case Diagram of Proposed System

3.2.3 Class Diagram of Proposed System

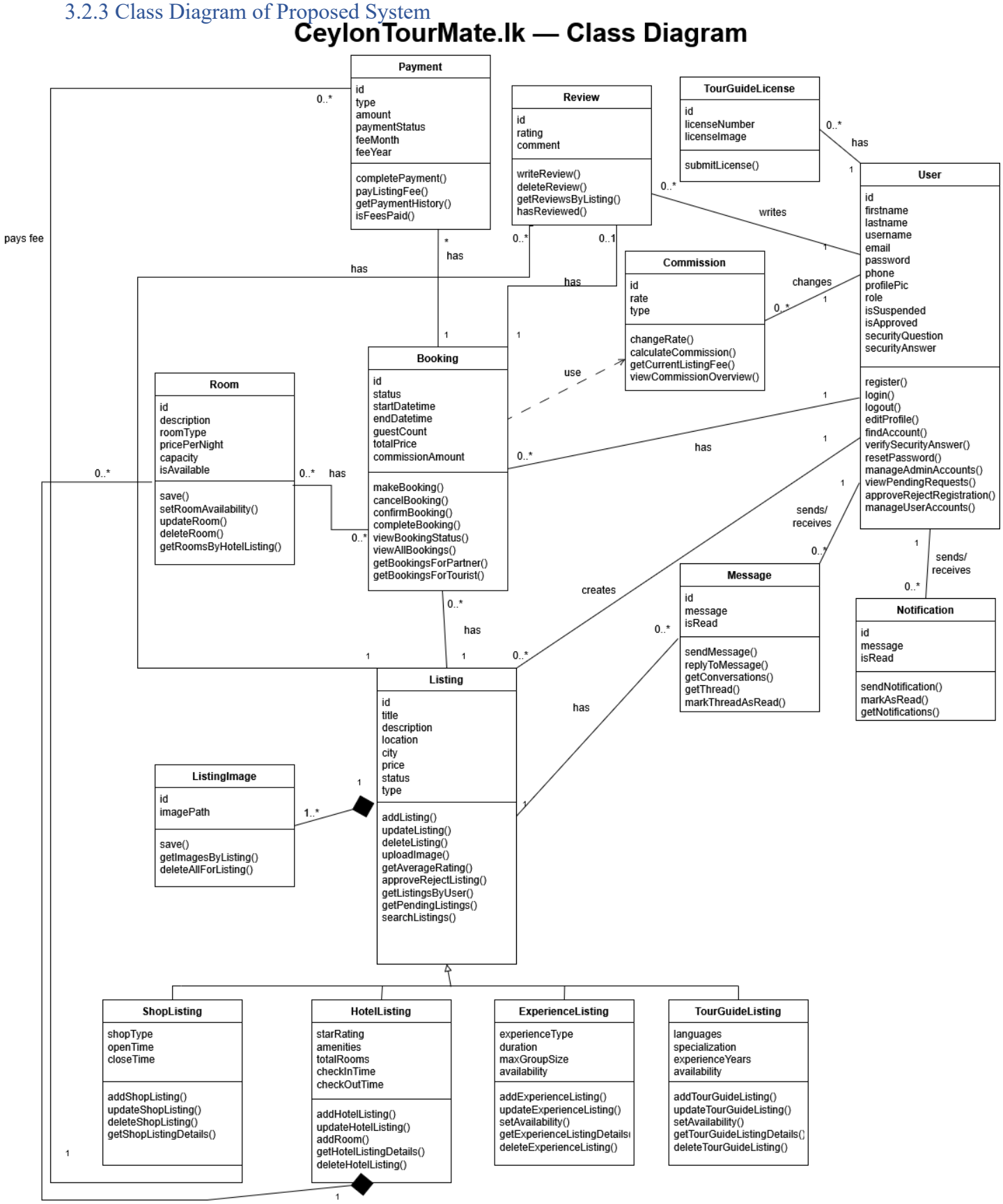


Figure 3 Class Diagram of Proposed System

3.2.4 Sequence Diagrams of Proposed System

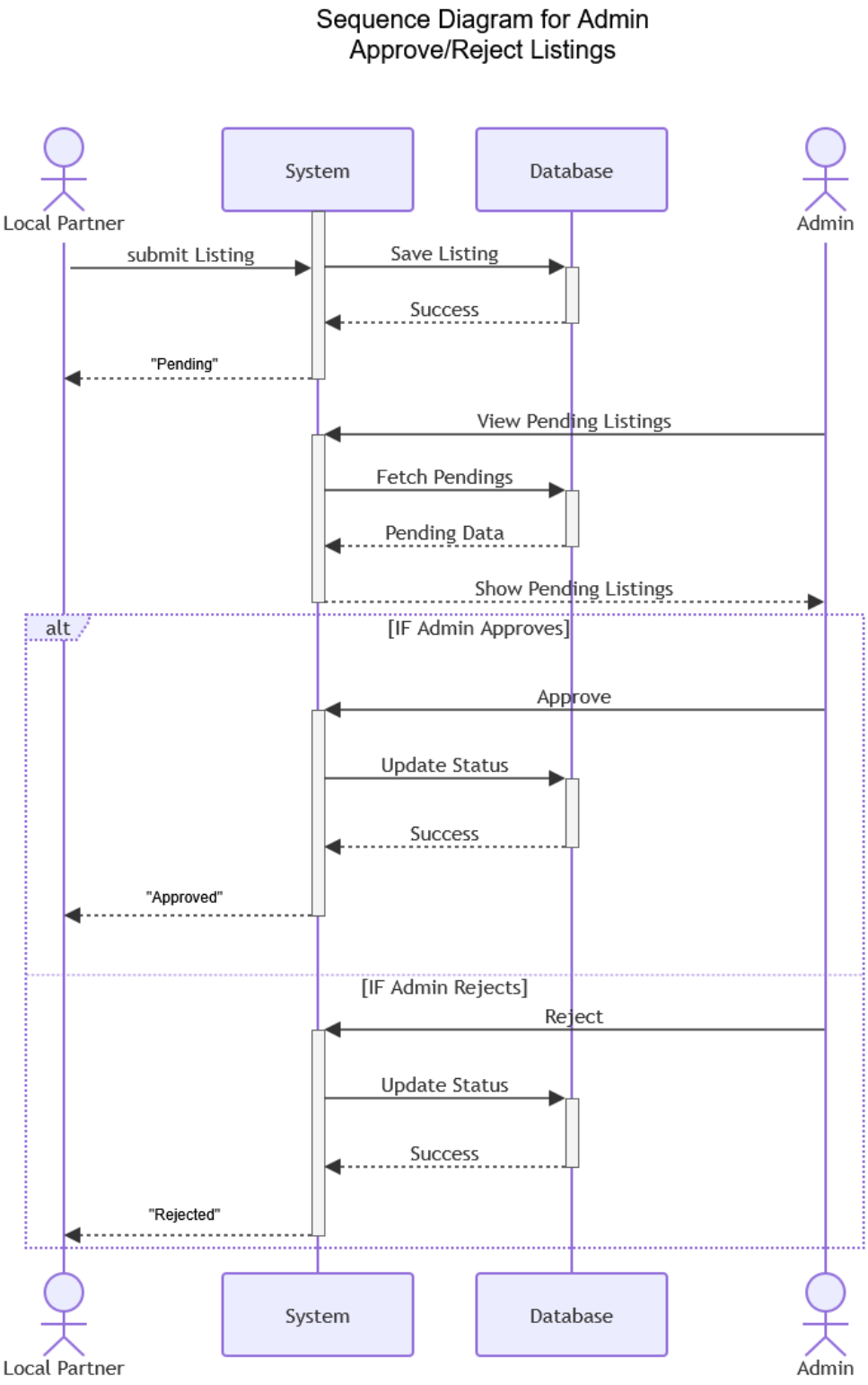


Figure 4 Sequence Diagram for Admin Approve/Reject Listings

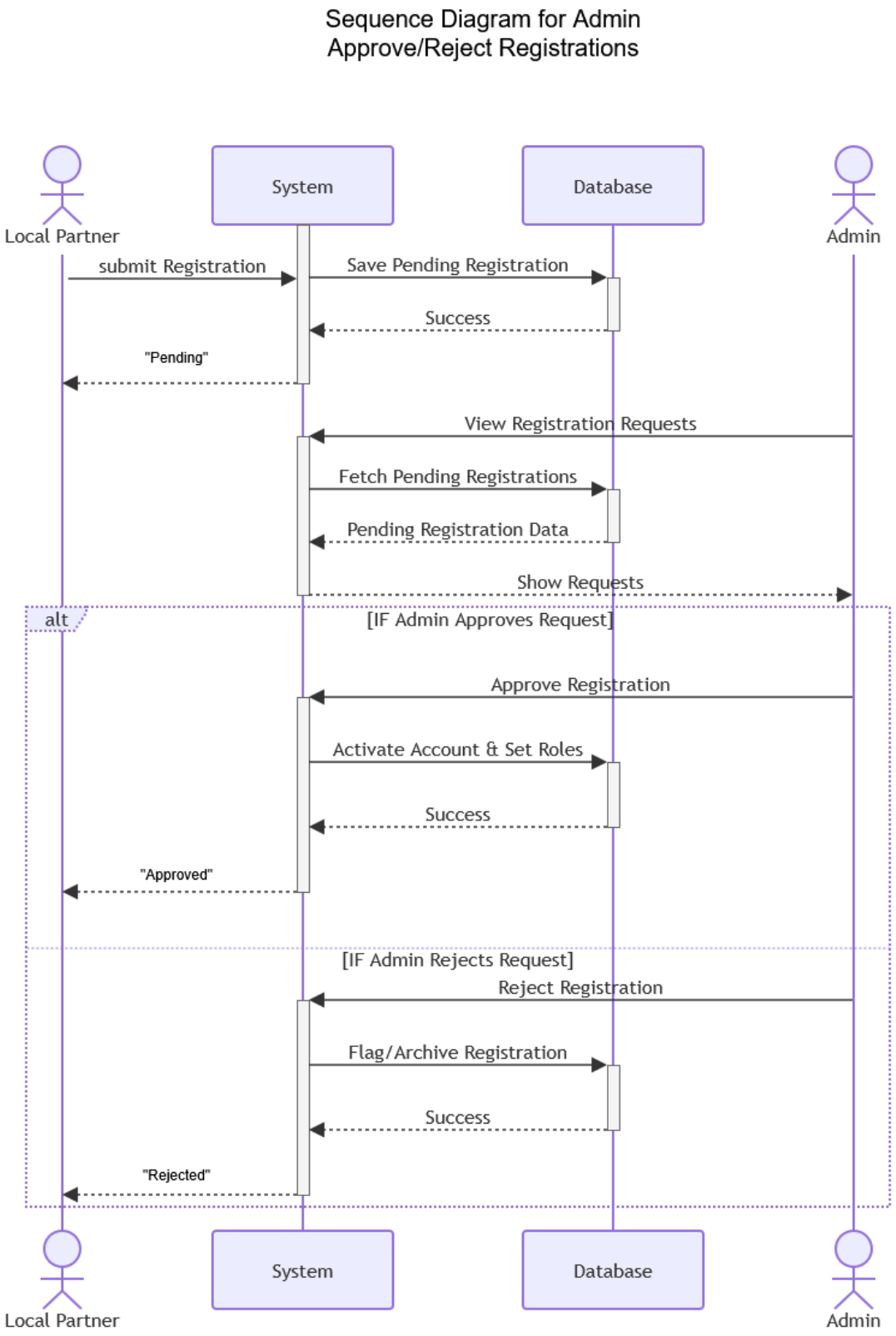


Figure 5 Sequence Diagram for Admin Approve/Reject Registrations

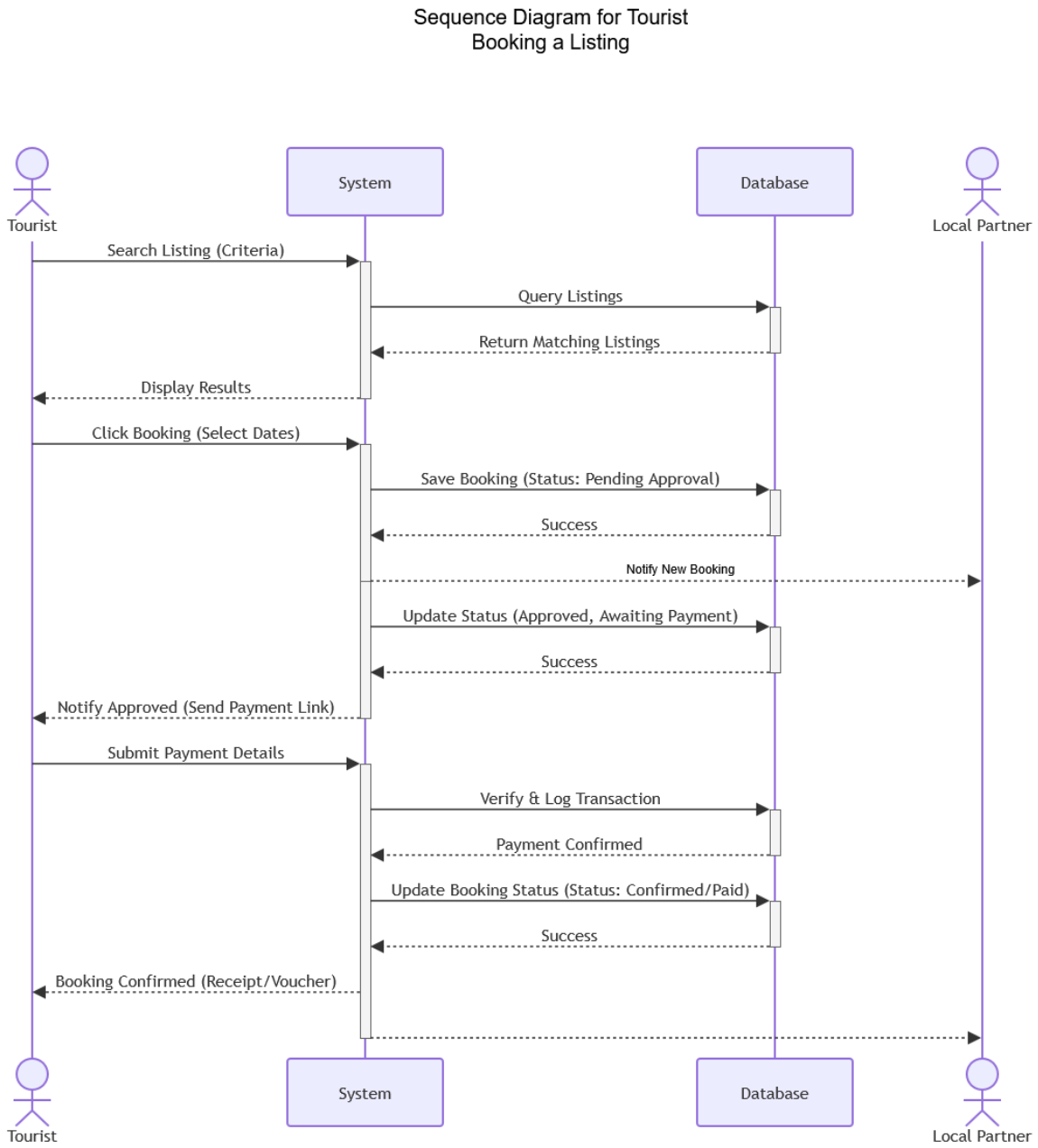


Figure 6 Sequence Diagram for Tourist Booking a Listing

3.3 ER Diagram of Proposed System

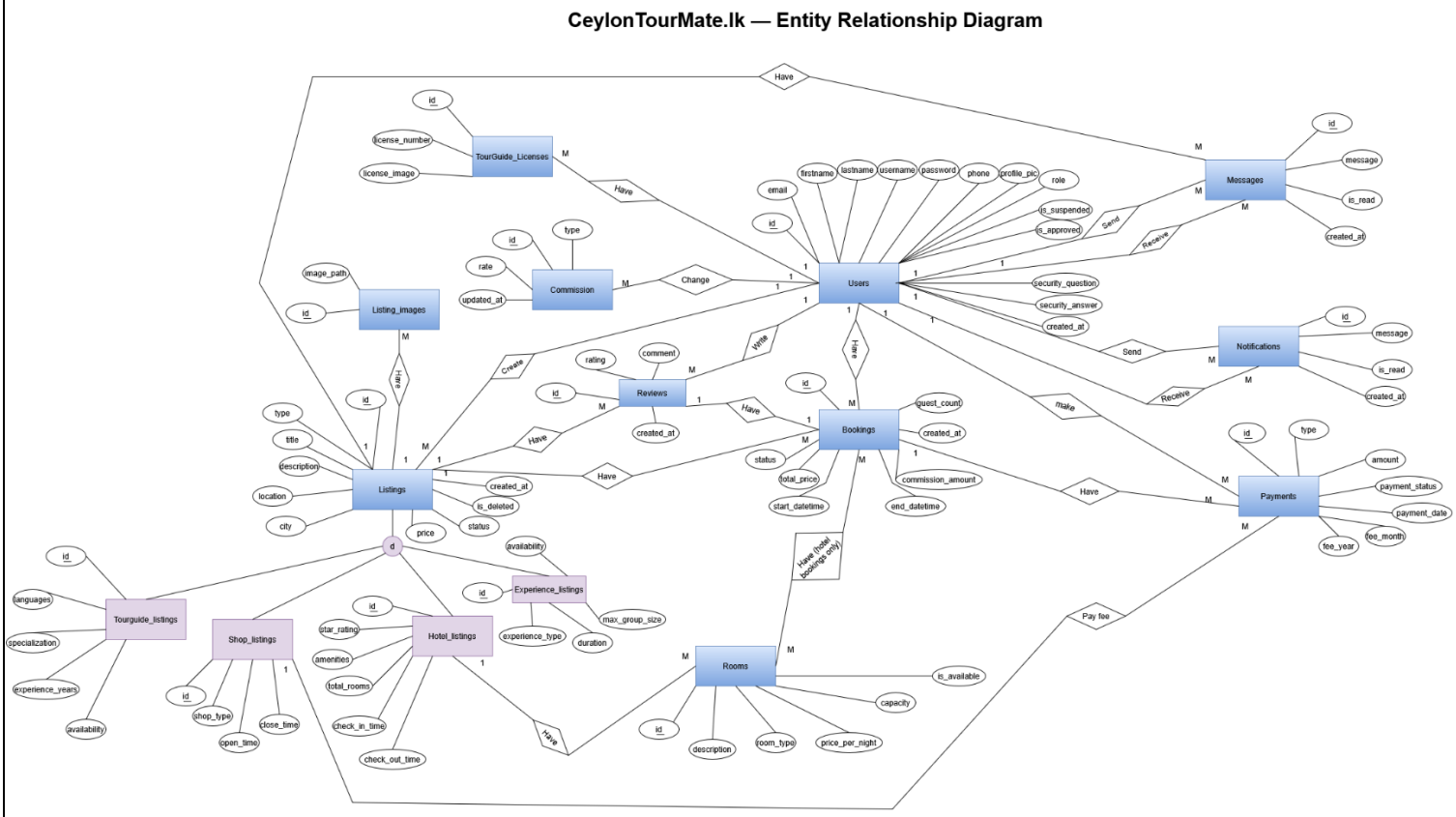


Figure 7 ER Diagram of Proposed System

3.4 Chapter Summery

In this chapter I presented all of my diagrams including usecase, class, sequence and ER. These represent the analysis and modeling stage my CeylonTourMatelk.Use case diagram describes the system actors and their interactions, class diagram describes system's OOP classes, sequence diagram describes the flow of each use case and finally ER diagram describes the database design.

Chapter 4: Solution Design

4.1 Introduction

In this chapter im going to present the design of the proposed solution for CeylonTourMateLK, which include all interface designs, database design, and the report layouts.

4.2 Interface Design

- INT-01 Home Page – Landing page of the platform

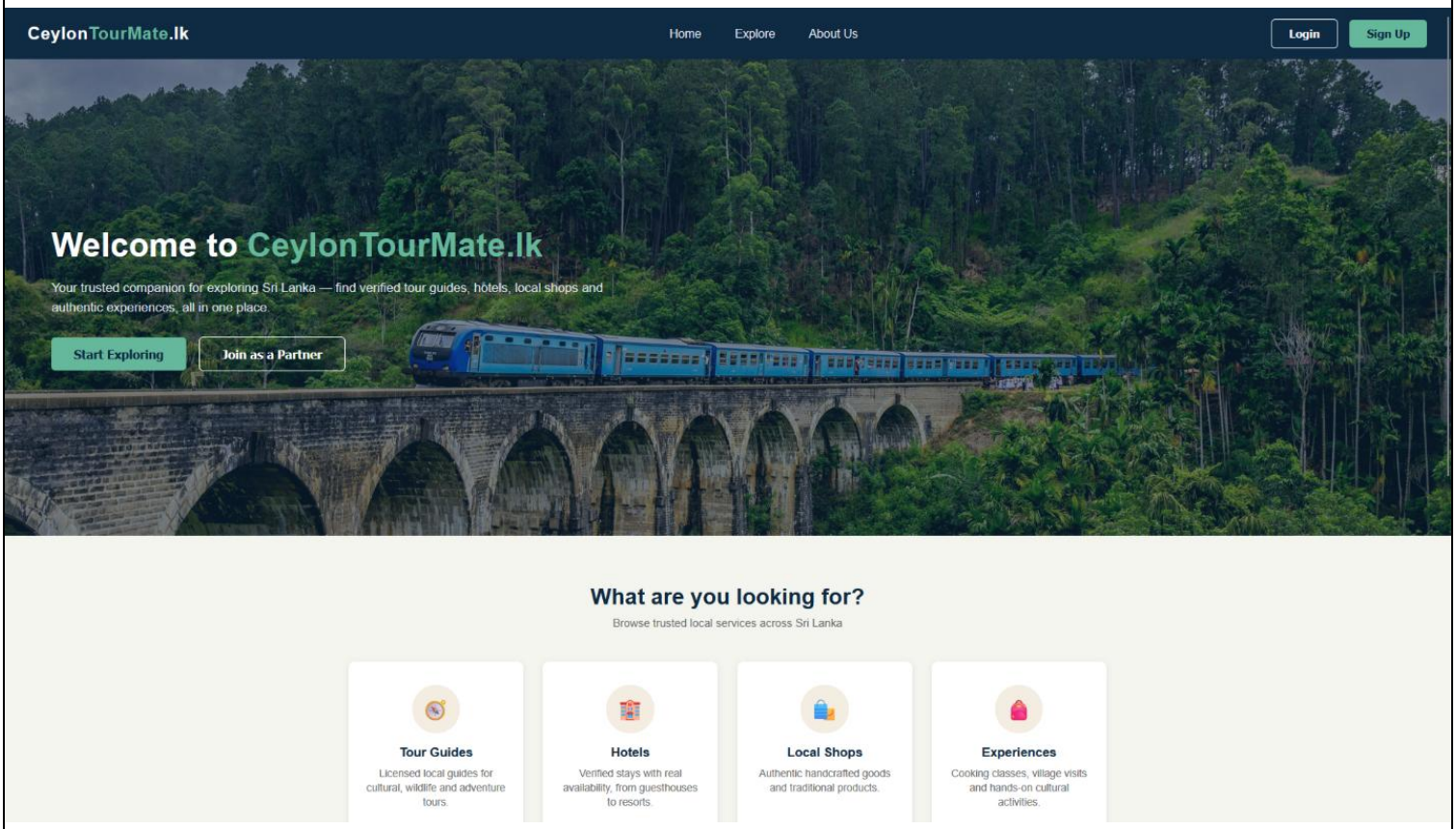


Figure 8 Interface 1

- INT-02 Registration Page – allow all users to register in the platforms, for local partners admin approval required.

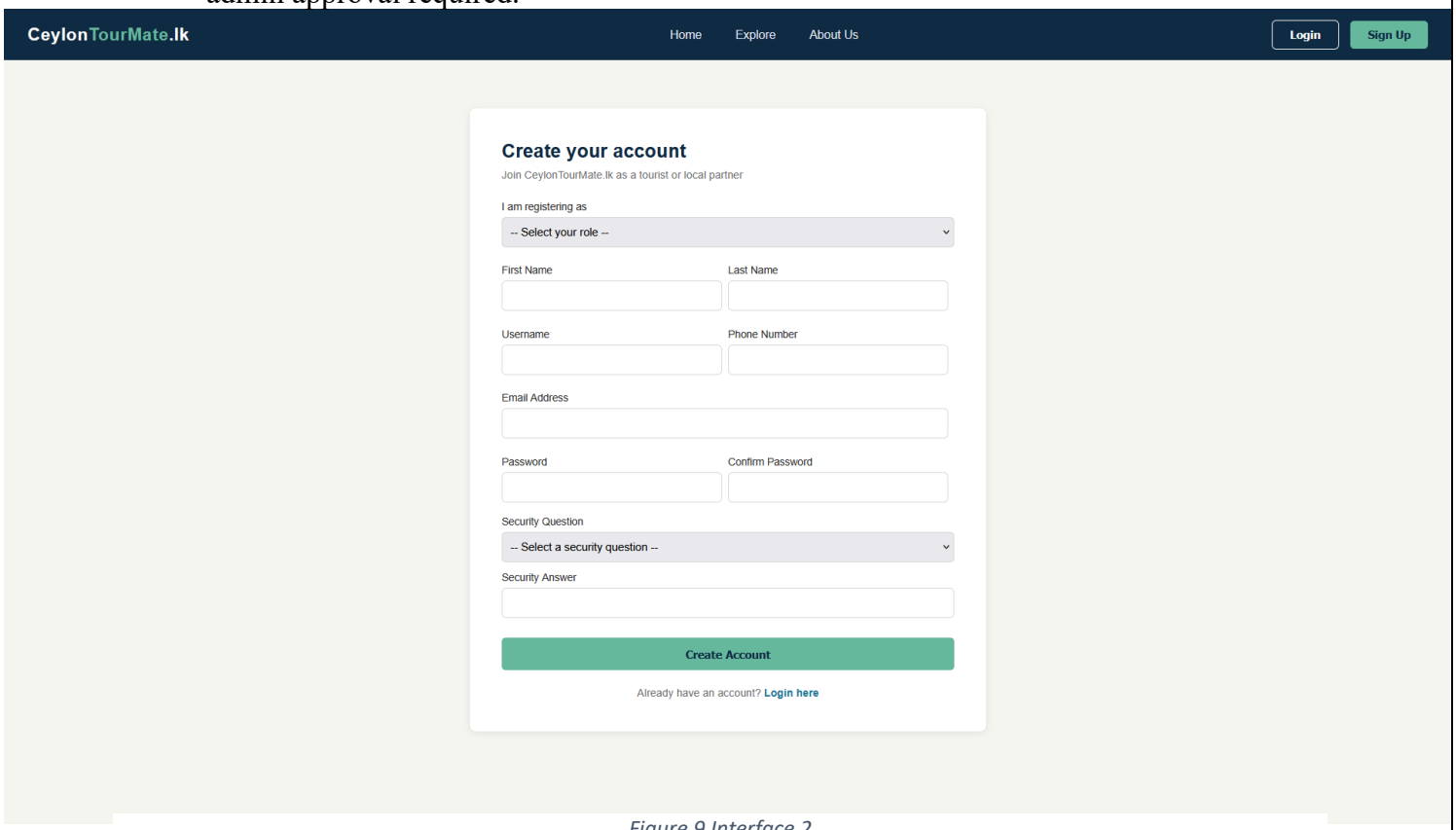


Figure 9 Interface 2

- INT-03 Login page – allows users to login to the system, role based login.

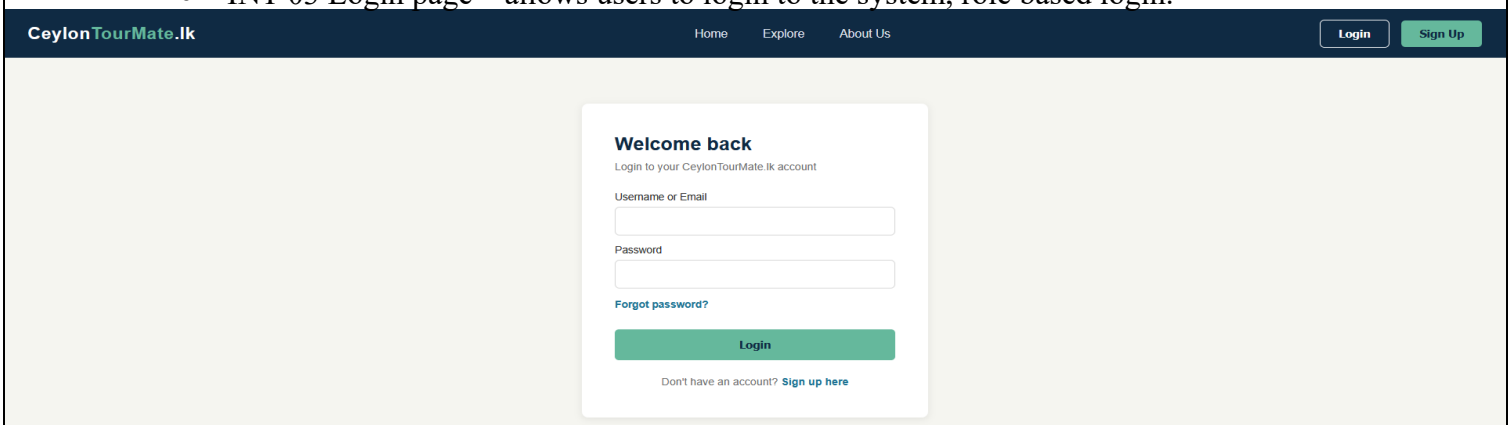


Figure 10 Interface 3

- INT-04 Admin dashboard – where admin can see his dashboard and other controls.

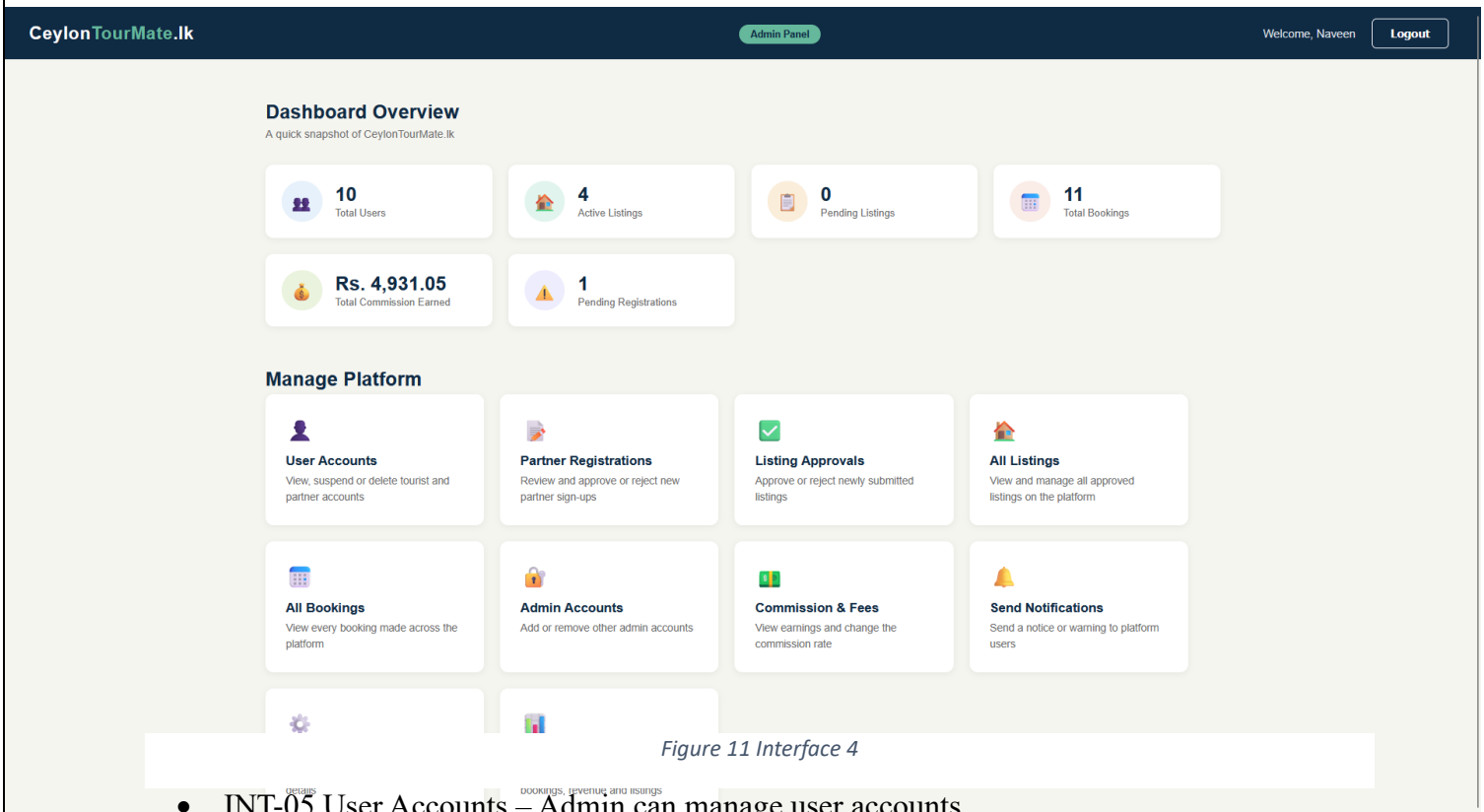


Figure 11 Interface 4

- INT-05 User Accounts – Admin can manage user accounts

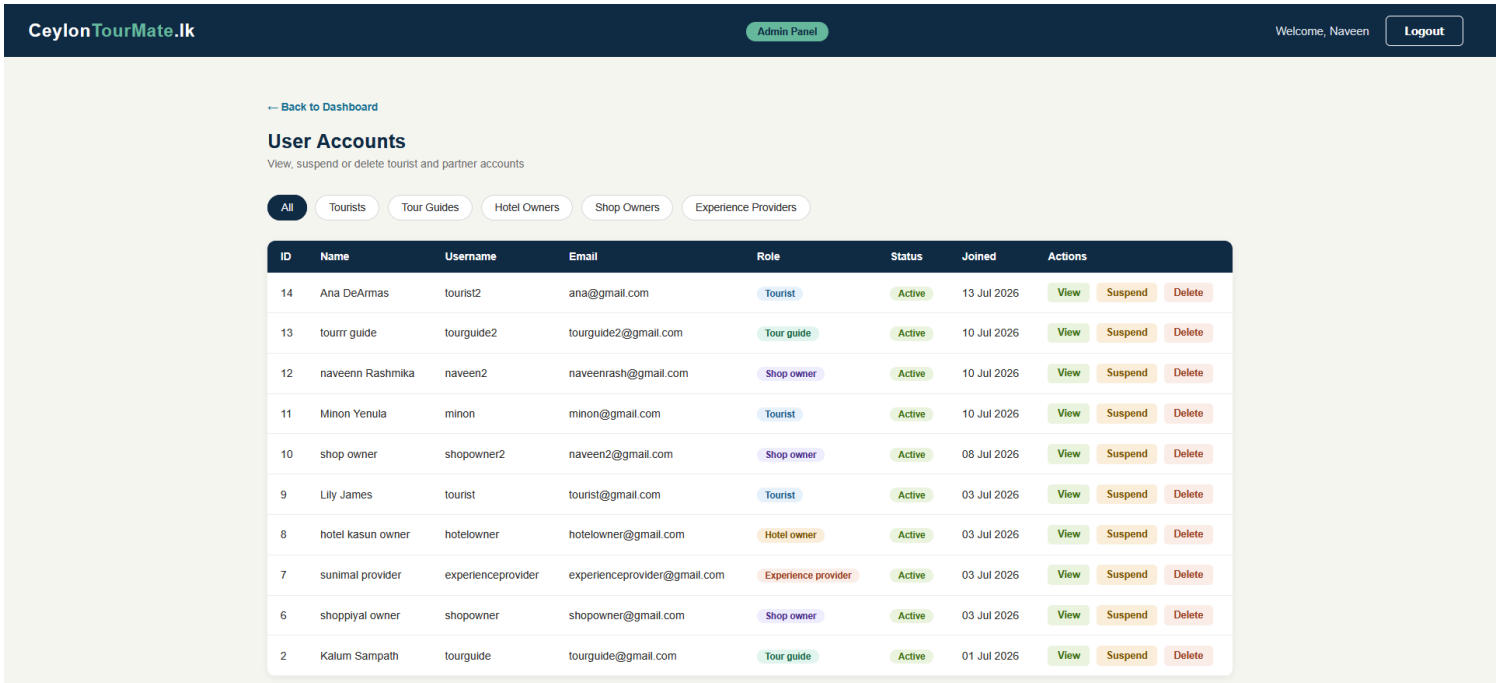


Figure 12 Interface 5

- INT-06 Parter registrations – admin can manage parter registrations

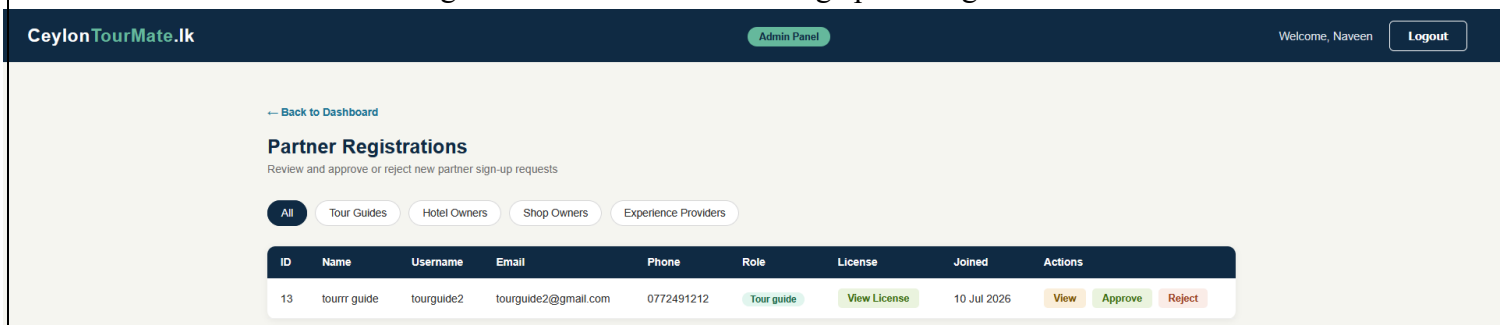


Figure 13 Interface 6

- INT -07 Listing approvals – admin can manage partner listings

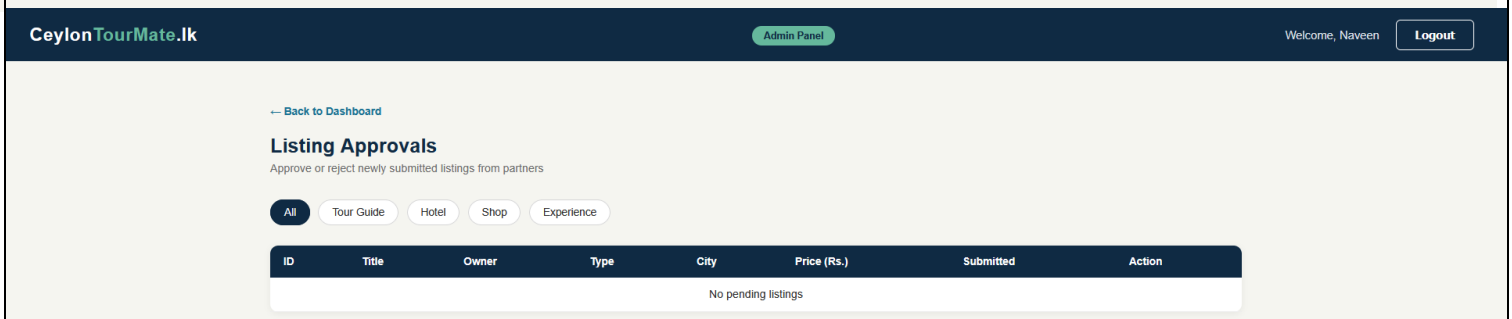


Figure 14 Interface 7

- INT -08 All listings – admin can manage partner listings

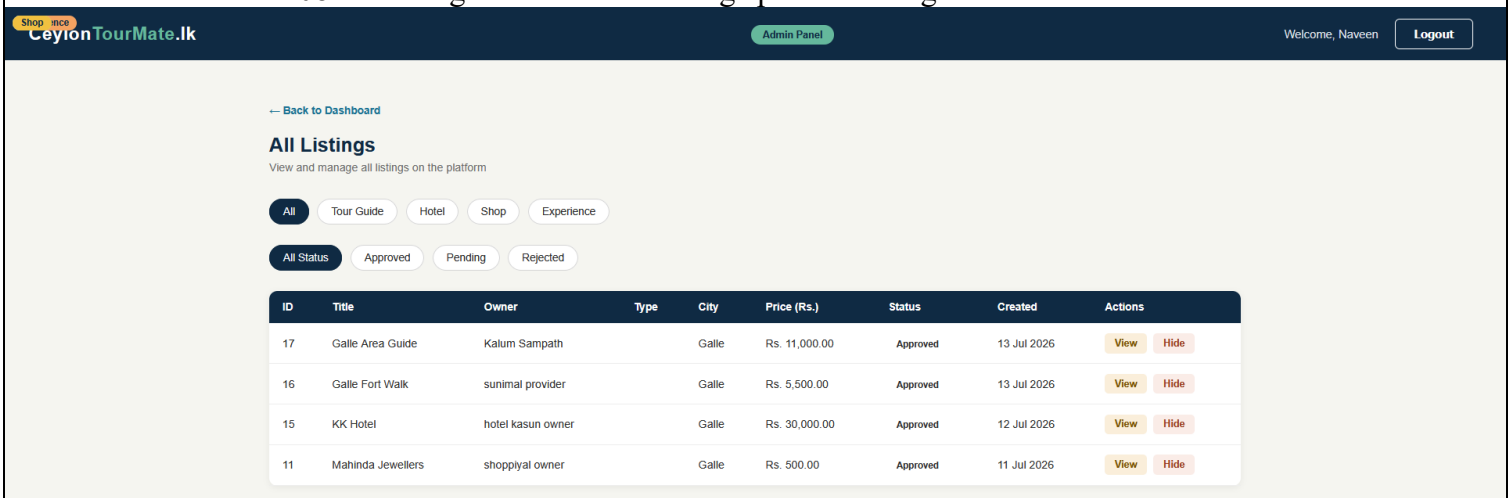


Figure 16 Interface 8

- INT -09 All bookings – admin can view all bookings in the system

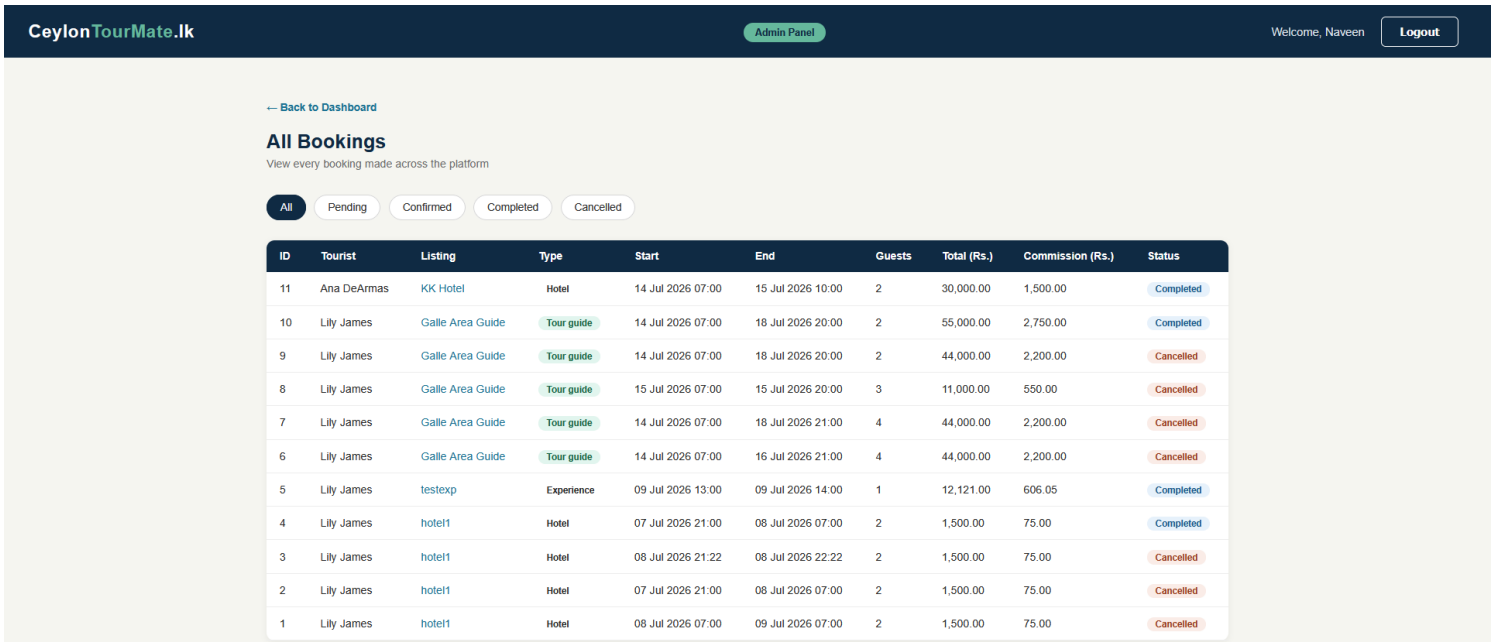


Figure 15 Interface 9

- INT -10 Admin accounts -admin can manage admin accounts

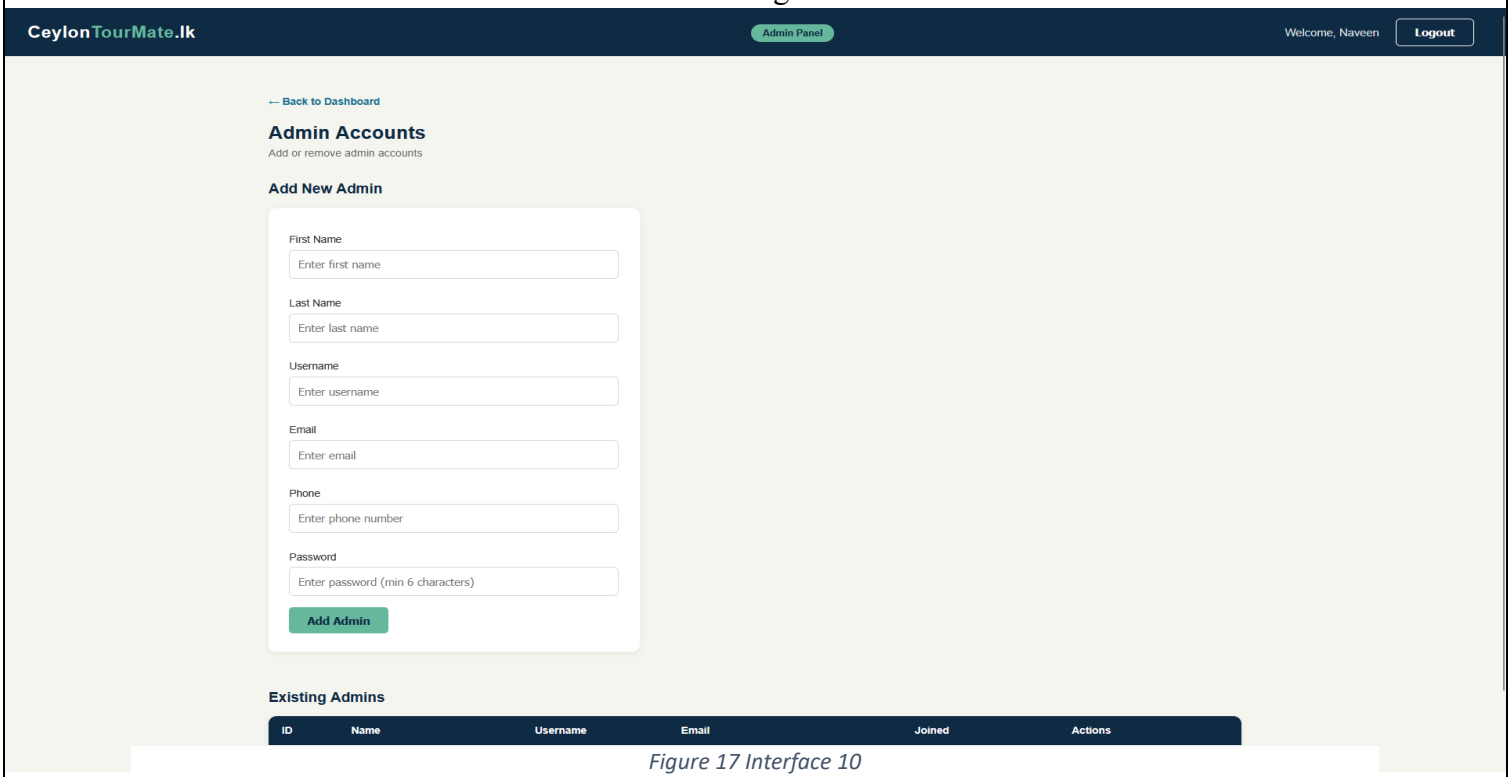


Figure 17 Interface 10

- INT -11 Commission fee overview – admin can view current com rates and manage them

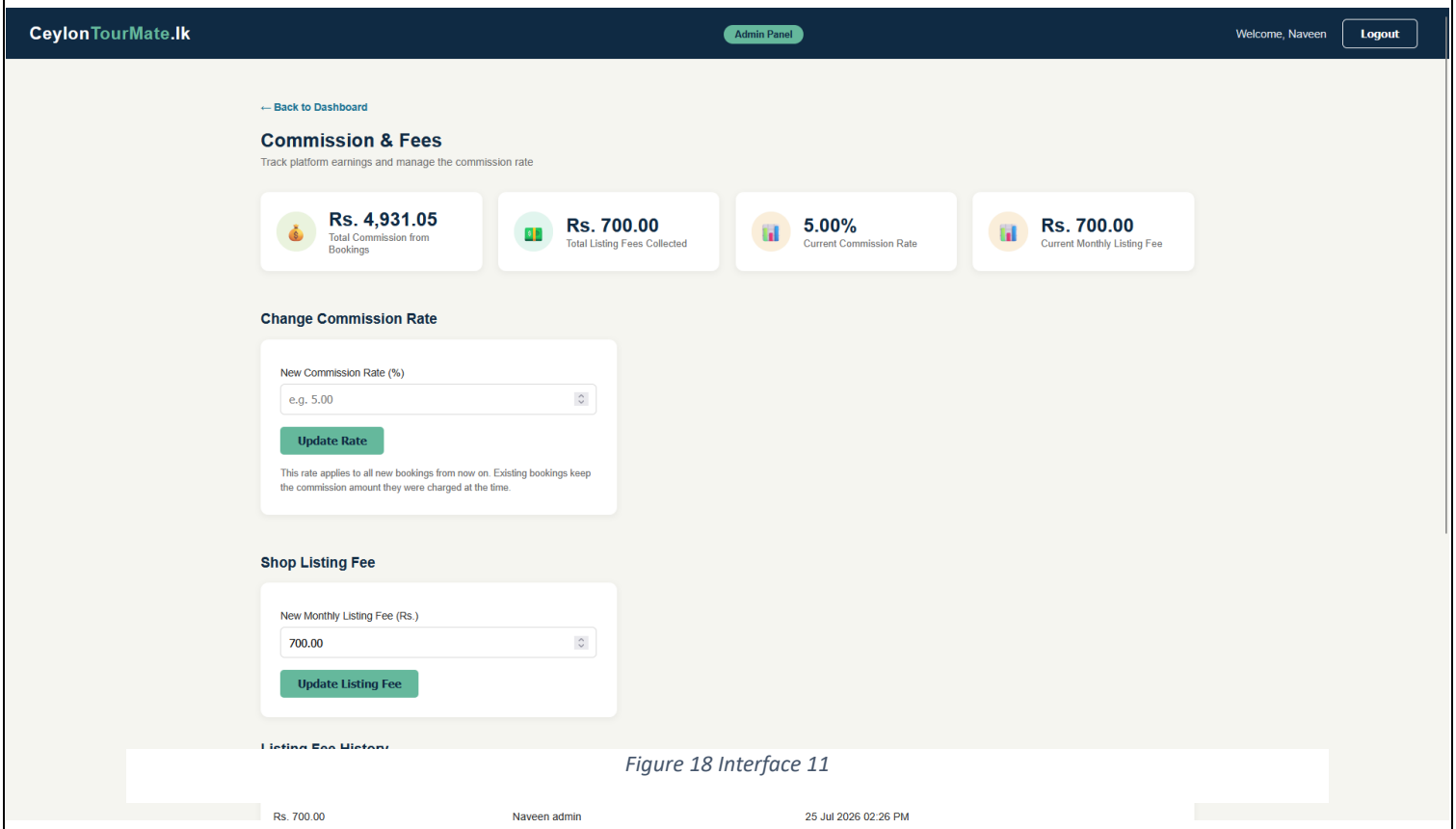


Figure 18 Interface 11

- INT -12 Send notification – Admin can send notifications to users

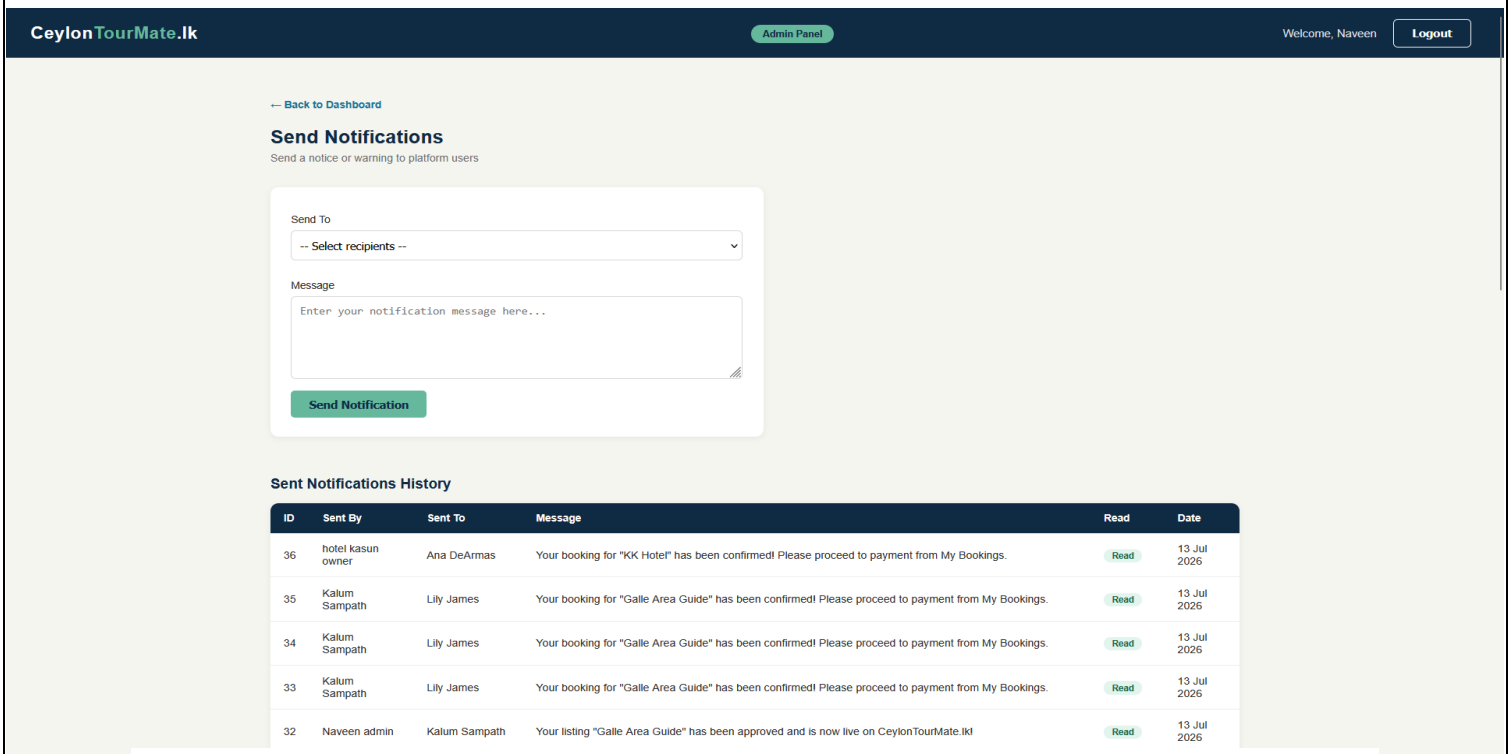


Figure 19 Interface 12

- INT -13 My profile – users can change their profile (same interface for everyone)

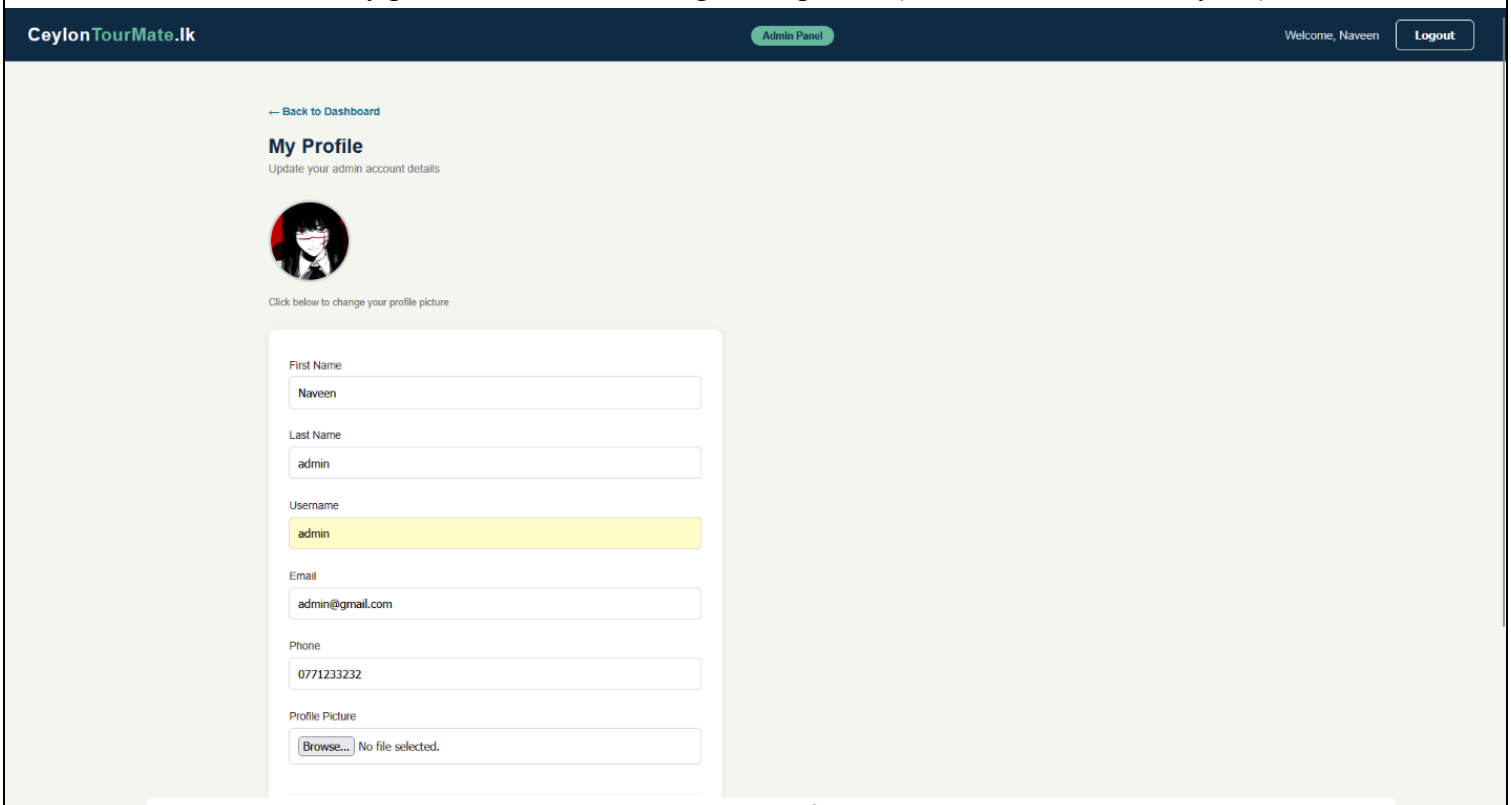


Figure 20 Interface 13

- INT -14 Reports – Admin can generate reports

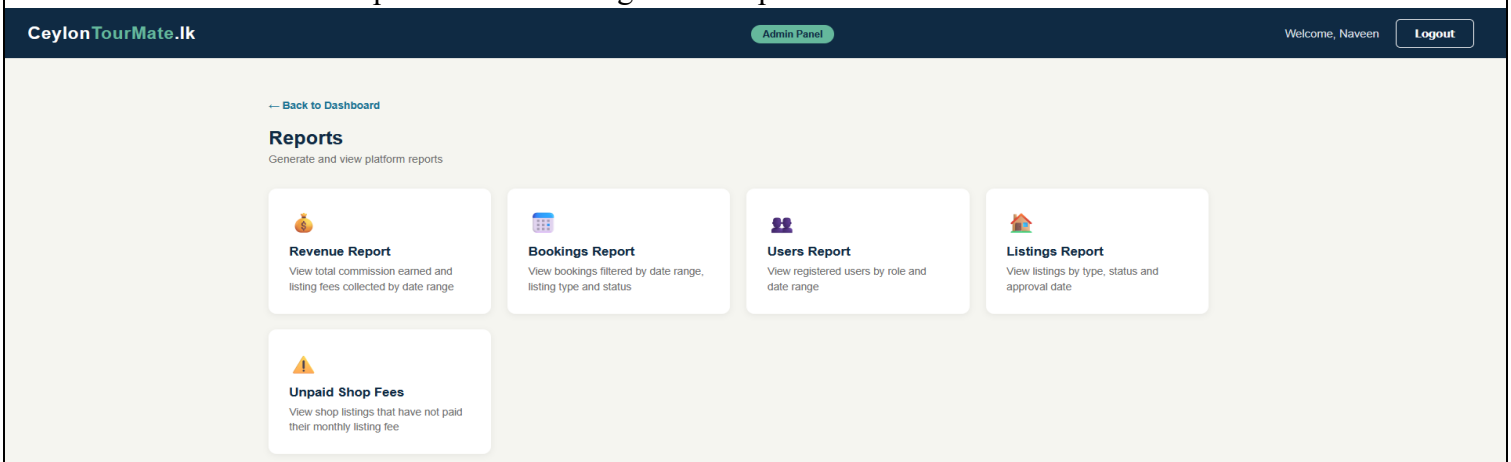


Figure 21 Interface 14

- INT -14 Experience provider dashboard

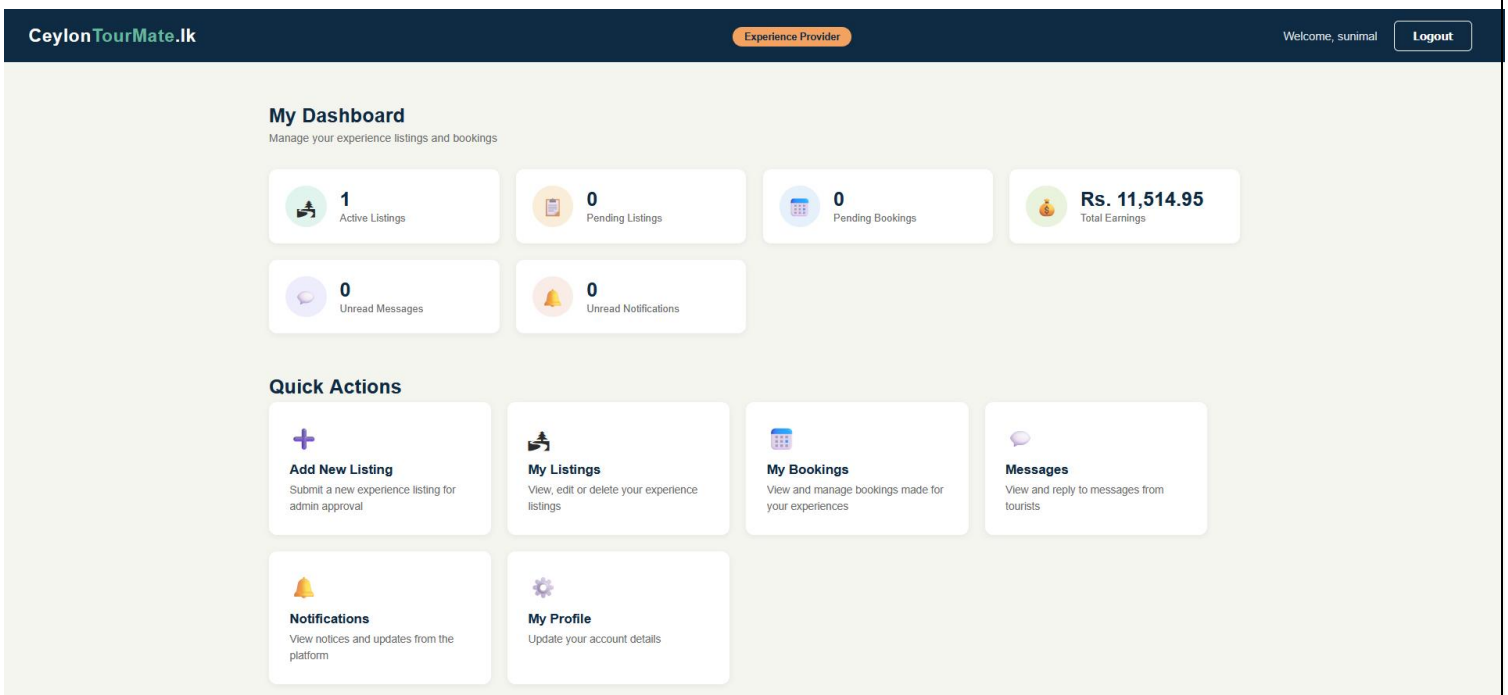


Figure 22 Interface 15

- INT -15 Hotel owner dashboard

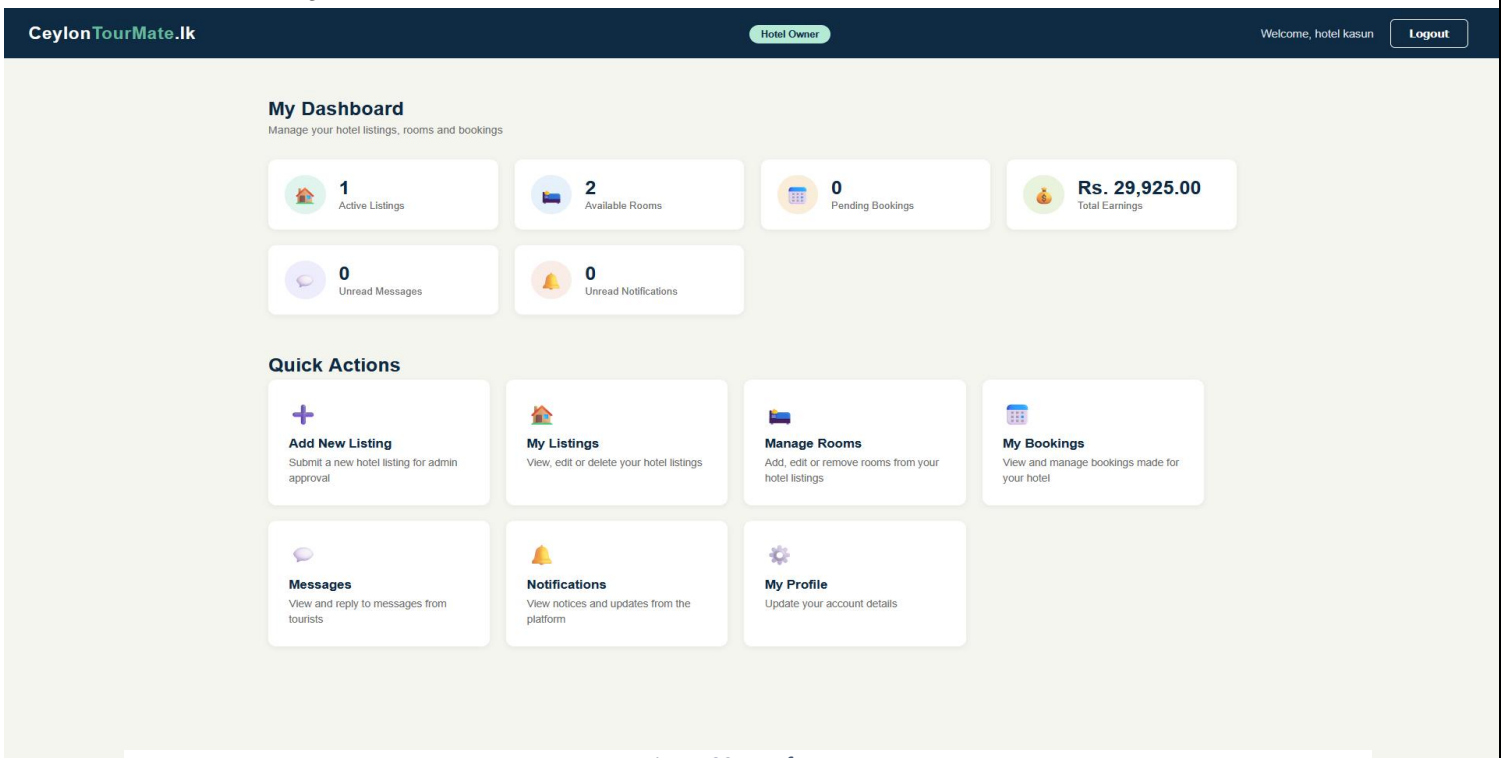


Figure 23 Interface 15

- INT -16 Shop owner dashboard

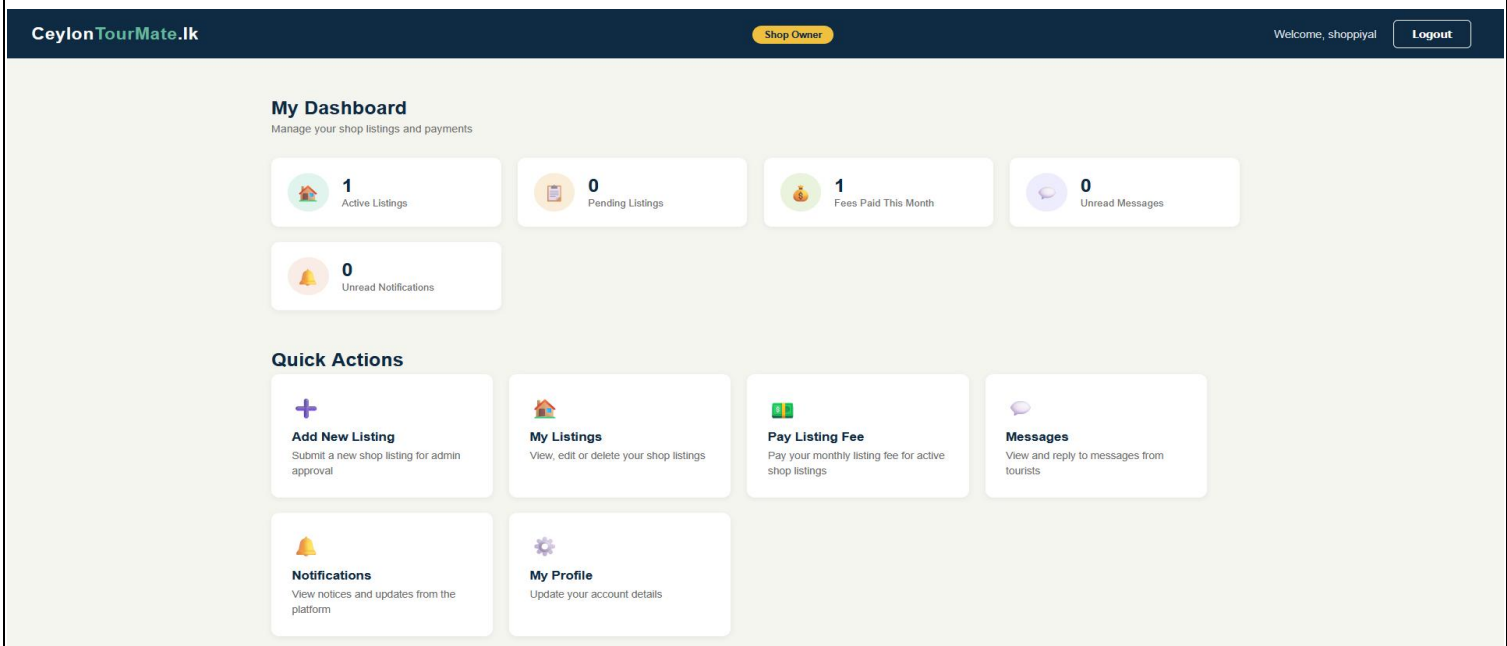


Figure 24 Interface 16

- INT -17 Tour guide dashboard

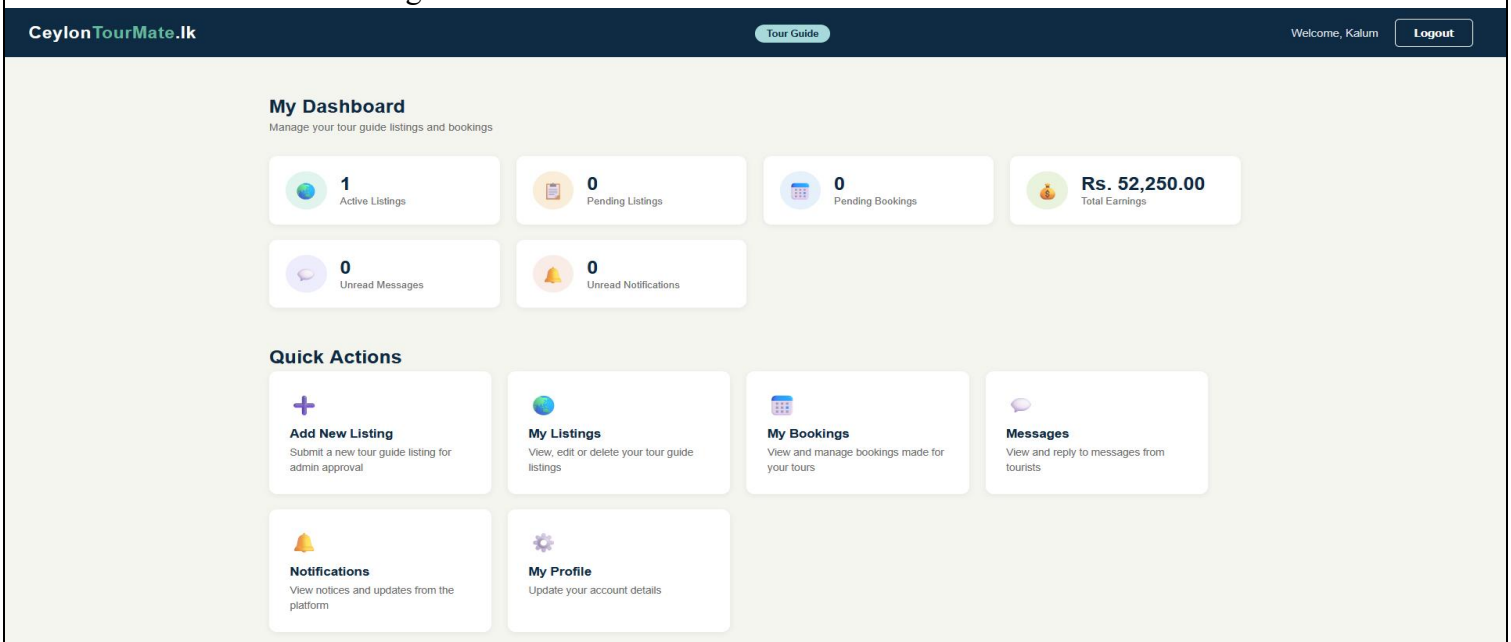


Figure 26 Interface 17

- INT -18 Tourist dashboard

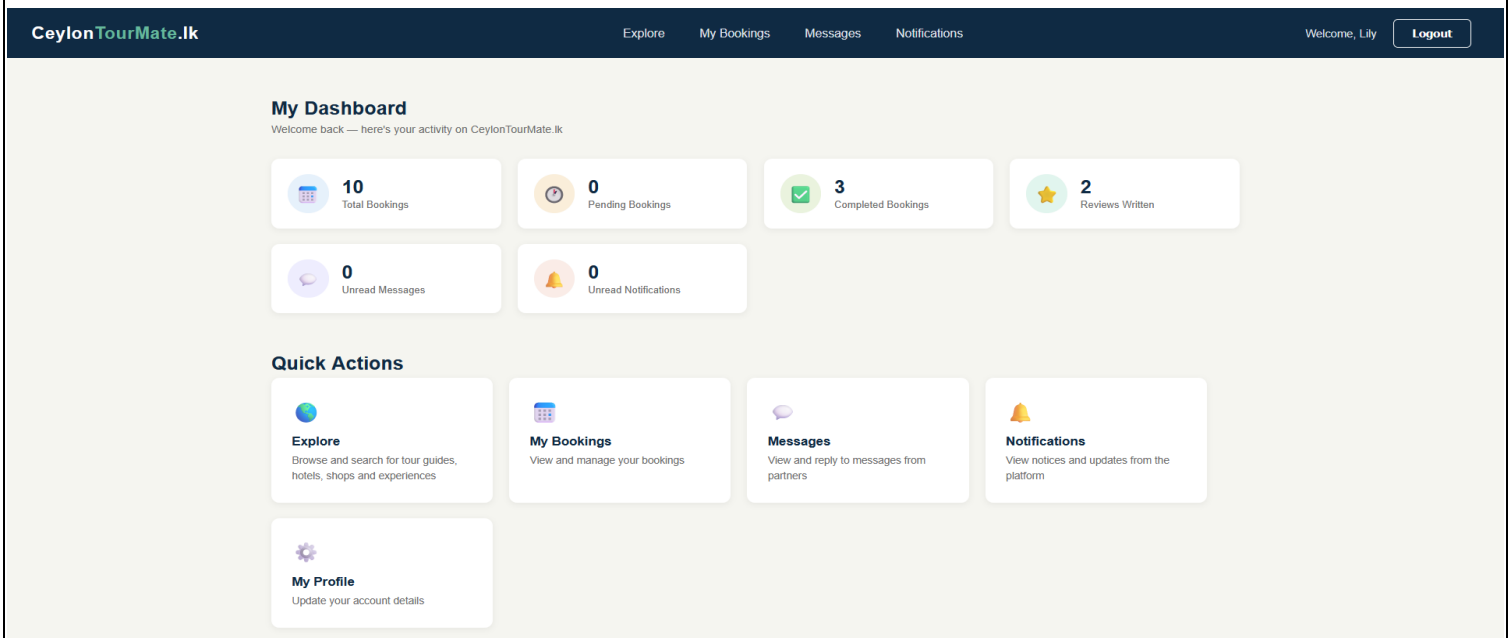


Figure 25 Interface 18

- INT -19 Explorer page – Tourist can explore services

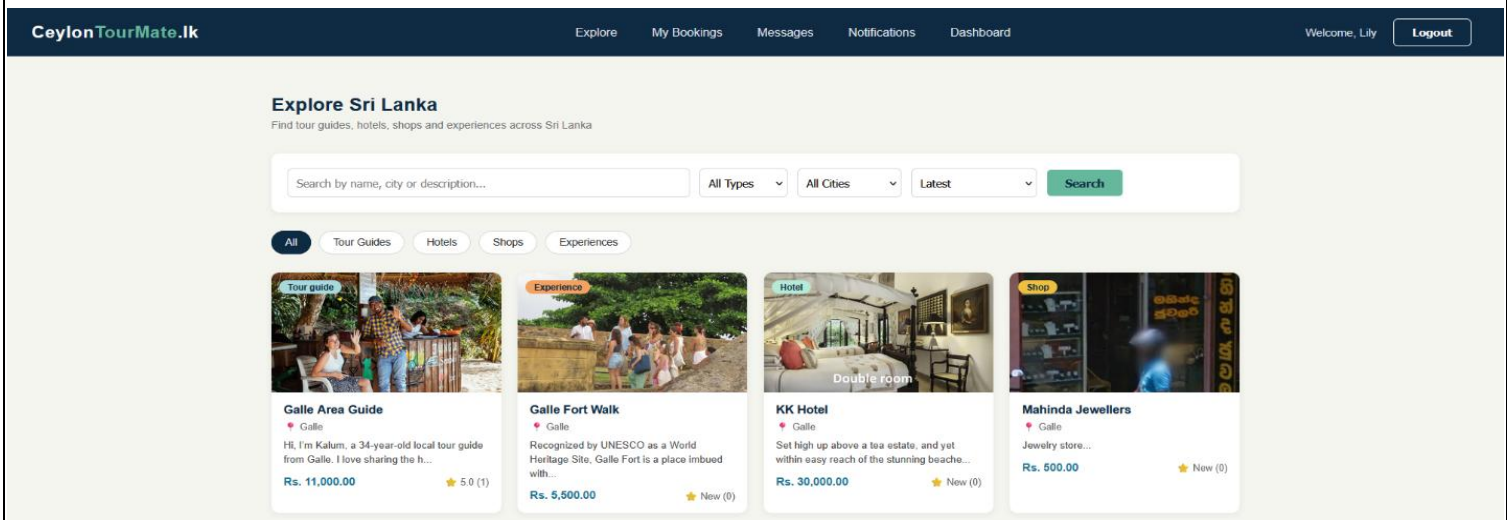


Figure 27 Interface 19

- INT -20 Tourist booking page

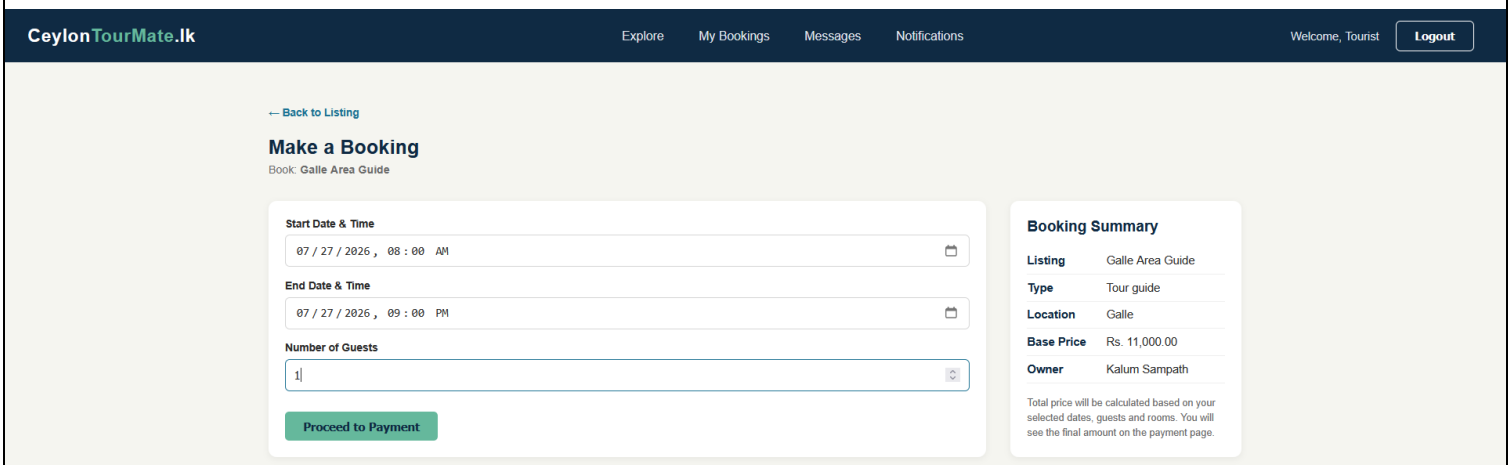


Figure 28 Interface 20

4.3 Database Design

Table 1:Users

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID for each user (Primary Key)
firstname	Text	50	User's first name
lastname	Text	50	User's last name
username	Text	50	Must be unique
email	Text	100	Must be unique
password	Text	255	Encrypted password
phone	Text	15	Contact number
profile_pic	Text	255	File path to profile picture (default: default.png)
role	Text (fixed options)	1	Options: tourist, tour_guide, shop_owner, hotel_owner, experience_provider, admin
is_suspended	Yes/No	1	Whether the account is suspended by admin
is_approved	Yes/No	1	Whether the account has been approved
security_question	Text	10	Used for password recovery
security_answer	Text	255	Answer to security question
created_at	DateTime	4	Date and time account was created

Table 2:Listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID for each listing (Primary Key)
user_id	Number	100	Foreign Key → links to Users table (owner of listing)
title	Text	255	Listing title
description	Text (Long)	Variable	Full description of the listing
location	Text	255	Specific address/area
city	Text	100	City name
price	Currency	10	Base price of listing
status	Text (fixed options)	1	Options: pending, approved, rejected
type	Text (fixed options)	1	Options: tour_guide, shop, hotel, experience
created_at	DateTime	4	Date and time listing was created
is_deleted	Yes/No	1	Soft-delete flag (1 = hidden, not permanently removed)

Table 3:listing_images

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID for each image (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
image_path	Text	255	File path to image, not the image itself

Table 4:hotel_listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
star_rating	Number	4	Hotel rating from 1 to 5 stars
amenities	Text (Long)	100	Comma-separated list of hotel amenities
total_rooms	Number	4	Total number of rooms in the hotel
check_in_time	DateTime (Time only)	10	Standard check-in time
check_out_time	DateTime (Time only)	10	Standard check-out time

Table 5:rooms

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
hotel_listing_id	Number	4	Foreign Key → links to Hotel Listings table
description	Text (Long)	100	Room description
room_type	Text (fixed options)	1	Options: single, double, suite, family
price_per_night	Currency	5	Price for one night
capacity	Number	4	Maximum number of guests
is_available	Yes/No	1	Whether the room is currently available

Table 6:shop_listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
shop_type	Text	100	e.g. Jewellery, Souvenir
open_time	DateTime (Time only)	10	Shop opening time
close_time	DateTime (Time only)	10	Shop closing time

Table 7:experience_listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
experience_type	Text	100	e.g. Historical, Photography
duration	Text	100	e.g. "2 hours"
max_group_size	Number	4	Maximum people allowed per group
availability	Text	255	e.g. "Daily", "Weekends only"

Table 8:tourguide_listings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
listing_id	Number	4	Foreign Key → links to Listings table
languages	Text	255	Languages spoken by the guide
specialization	Text	255	e.g. Historical, Adventure
experience_years	Number	4	Years of guiding experience
availability	Text	255	e.g. "Daily", "Weekdays only"

Table 9:tourguide_license

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
user_id	Number	4	Foreign Key → links to Users table
license_number	Text	100	Official tour guide license number
license_image	Text	255	File path to uploaded license image

Table 10:bookings

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
tourist_id	Number	4	Foreign Key → links to Users table (tourist who booked)
listing_id	Number	4	Foreign Key → links to Listings table
status	Text (fixed options)	1	Options: pending, confirmed, cancelled, completed
start_datetime	DateTime	10	Booking start date and time
end_datetime	DateTime	10	Booking end date and time
guest_count	Number	4	Number of guests included in booking
total_price	Currency	5	Total price of the booking
commission_amount	Currency	5	Platform's commission earned from this booking
created_at	DateTime	10	Date and time booking was made

Table 11:booking_rooms

Field Name	Data Type	Size (bytes)	Remarks
booking_id	Number	4	Foreign Key → links to Bookings table (part of Composite Primary Key)
room_id	Number	4	Foreign Key → links to Rooms table (part of Composite Primary Key)

Table 12:Payments

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
user_id	Number	4	Foreign Key → links to Users table
shop_listing_id	Number	4	Foreign Key → links to Shop Listings table (used for listing fee payments)
booking_id	Number	4	Foreign Key → links to Bookings table (used for booking payments)
type	Text (fixed options)	1	Options: booking, listing_fee
amount	Currency	5	Payment amount
payment_status	Text (fixed options)	1	Options: pending, completed, failed
fee_month	Number	4	Month number, used only for listing fee payments
fee_year	Number	4	Year, used only for listing fee payments
payment_date	DateTime	10	Date and time payment was made

Table 13:reviews

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
tourist_id	Number	4	Foreign Key → links to Users table
listing_id	Number	4	Foreign Key → links to Listings table
booking_id	Number	4	Foreign Key → links to Bookings table
rating	Number	4	Rating score, restricted to 1–5
comment	Text (Long)	100	Reviewer's written comment
created_at	DateTime	10	Date and time review was posted

Table 13:messages

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
sender_id	Number	4	Foreign Key → links to Users table
receiver_id	Number	4	Foreign Key → links to Users table
listing_id	Number	4	Foreign Key → links to Listings table (message relates to this listing)
message	Text (Long)	100	Message content
is_read	Yes/No	1	Whether the message has been read
created_at	DateTime	10	Date and time message was sent

Table 14:notifications

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
sender_id	Number	4	Foreign Key → links to Users table (usually admin/system)
receiver_id	Number	4	Foreign Key → links to Users table
message	Text (Long)	100	Notification content
is_read	Yes/No	1	Whether the notification has been read
created_at	DateTime	10	Date and time notification was created

Table 15:commision

Field Name	Data Type	Size (bytes)	Remarks
id	Number	4	Unique ID (Primary Key)
rate	Currency	3	Commission rate value (percentage or fixed fee)
type	Text (fixed options)	1	Options: booking_commission, listing_fee
updated_by	Number	4	Foreign Key → links to Users table (admin who made the change)
updated_at	DateTime	10	Date and time rate was last updated

4.4 Report Layout Design

- RL-01 Shop listing report fee report – admin can view shop listings that have not paid the listing fee.



Figure 29 report layout 1

- RL-02 All listings report- admin can view all listing by type

CeylonTourMate.lk

All Listings Separated by Type

Generated on: 25 Jul 2026 08:46 PM

Listing Type	Total	Approved	Pending	Rejected	Avg Rating	Total Reviews
Tour guide	1	1	0	0	5.0	1
Shop	1	1	0	0	N/A	0
Hotel	1	1	0	0	N/A	0
Experience	1	1	0	0	N/A	0

Listings Summary

Total Listings on Platform 4

This report was generated automatically by CeylonTourMate.lk © 2026

Figure 30 report layout 2

- RL-03 All users reports – admin can view all user count on the platform by role

CeylonTourMate.lk

All Time User Statistics by Role

Generated on: 25 Jul 2026 08:46 PM

Role	Total	Active	Pending Approval	Suspended
Tourist	3	3	0	0
Tour guide	2	1	1	0
Shop owner	3	3	0	0
Hotel owner	1	1	0	0
Experience provider	1	1	0	0

Users Summary

Total Registered Users 10

This report was generated automatically by CeylonTourMate.lk © 2026

Figure 31 report layout 3

- RL-04 Monthly bookings report- admin can view all monthly bookings

CeylonTourMate.lk

Monthly Bookings by Listing Type

Generated on: 25 Jul 2026 08:46 PM

Month	Year	Type	Total	Completed	Confirmed	Pending	Cancelled	Revenue (Rs.)
July	2026	Tour guide	6	1	1	0	4	Rs. 209,000.00
July	2026	Hotel	5	2	0	0	3	Rs. 36,000.00
July	2026	Experience	1	1	0	0	0	Rs. 12,121.00

Bookings Summary

Total Bookings	**12**
Total Revenue Generated	**Rs. 257,121.00**

This report was generated automatically by CeylonTourMate.lk © 2026

Figure 32 report layout 4

- RL-05 Revenue report – admin can view the total platform's revenue.

CeylonTourMate.lk

Monthly Revenue — Commission & Listing Fees

Generated on: 25 Jul 2026 08:46 PM

Booking Commission Revenue

Month	Year	Completed Bookings	Commission Earned (Rs.)
July	2026	4	Rs. 4,931.05

Listing Fee Revenue

Month	Year	Payments Received	Fees Collected (Rs.)
July	2026	1	Rs. 700.00

Revenue Summary

Total Commission from Bookings	**Rs. 4,931.05**
Total Listing Fees Collected	**Rs. 700.00**
Grand Total Revenue	Rs. 5,631.05

This report was generated automatically by CeylonTourMate.lk © 2026

Figure 33 report layout 5

Chapter 5: Implementation

5.1 Introduction

In this chapter im going to explain the main implementation of my project, which covers system architecture, main classes and decisions I made during the coding part.

5.2 System architecture

As the architecture I used **3 tier architecture** for the development of CeylonTourMateLK with OOP phps and MYSQL database.

My main classes are as follows,

Booking.php	2026-07-15 8:21 PM	PHP Source File	13 KB
Commission.php	2026-07-15 8:31 PM	PHP Source File	5 KB
DBconnection.php	2026-07-15 8:28 PM	PHP Source File	1 KB
ExperienceListing.php	2026-07-15 8:28 PM	PHP Source File	5 KB
HotelListing.php	2026-07-15 8:28 PM	PHP Source File	5 KB
Listing.php	2026-07-15 8:28 PM	PHP Source File	10 KB
ListingImage.php	2026-07-15 8:28 PM	PHP Source File	2 KB
Message.php	2026-07-15 8:28 PM	PHP Source File	5 KB
Notification.php	2026-07-15 8:28 PM	PHP Source File	4 KB
Payment.php	2026-07-15 8:28 PM	PHP Source File	7 KB
Review.php	2026-07-15 8:28 PM	PHP Source File	4 KB
Room.php	2026-07-15 8:28 PM	PHP Source File	5 KB
session.php	2026-07-15 8:28 PM	PHP Source File	2 KB
ShopListing.php	2026-07-15 8:28 PM	PHP Source File	4 KB
TourGuideLicense.php	2026-07-15 8:28 PM	PHP Source File	2 KB
TourGuideListing.php	2026-07-15 8:28 PM	PHP Source File	5 KB
User.php	2026-07-15 8:28 PM	PHP Source File	12 KB

Figure 34 Main classes

5.3 Main Coding Areas

I used a session class to hold logged in users session information, which I use it to show a welcome message at the top of the platform.

```php
function checkSession() {
    if (session_status() == PHP_SESSION_NONE) {
        session_start();
    }
    if (!isset($_SESSION['user_id']) || !isset($_SESSION['role'])) {
        header("Location:login.php");
        exit();
    }
    if (!isset($_SESSION['firstname'])) {
        $conn = DBconnection::getInstance()->getConnection();
        $stmt = $conn->prepare("SELECT firstname FROM users WHERE id = ?");
        $stmt->bind_param("i", $_SESSION['user_id']);
        $stmt->execute();
        $result = $stmt->get_result();
        $user = $result->fetch_assoc();
        if ($user) {
            $_SESSION['firstname'] = $user['firstname'];
        }
    }
}
```

Figure 35 Session class

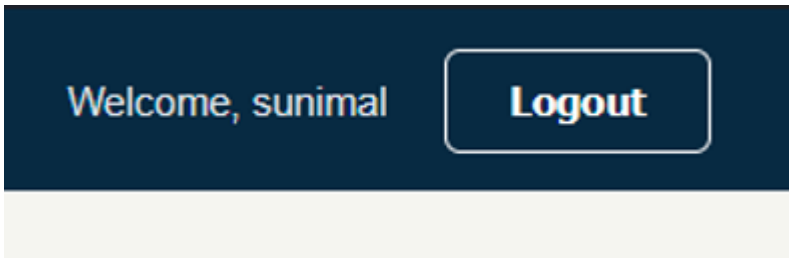
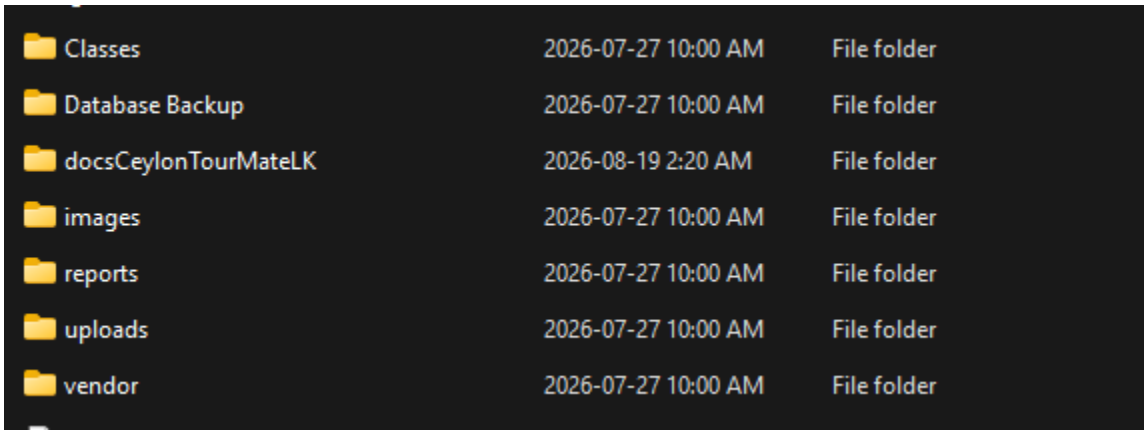
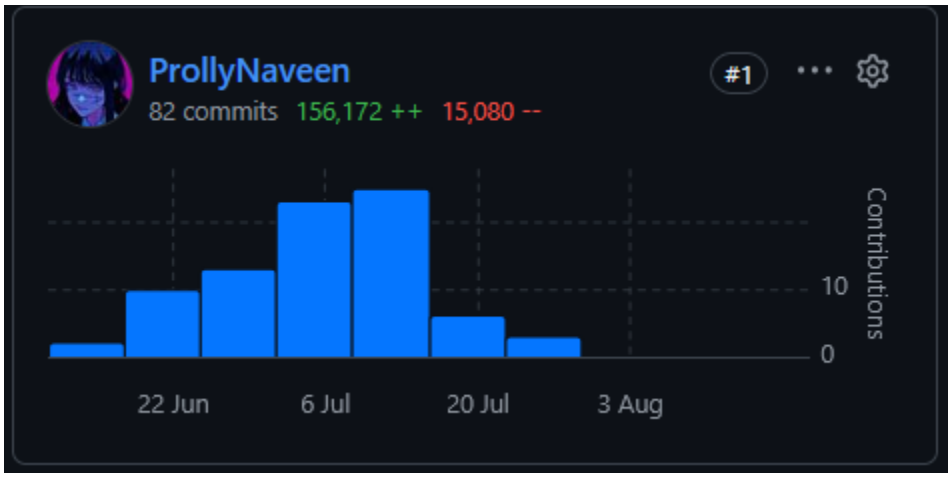


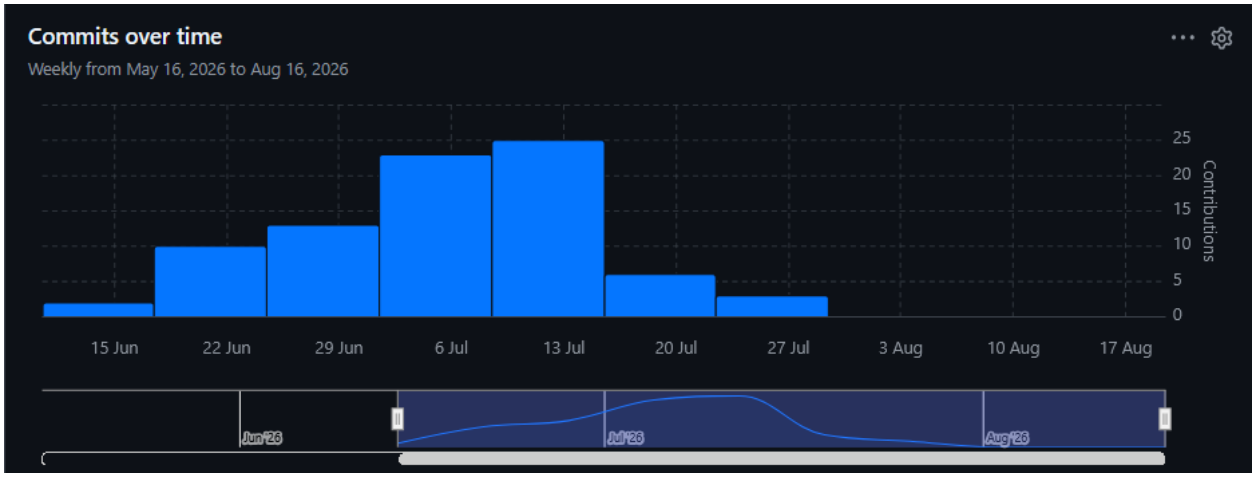
Figure 36 Welcome message

And for my whole project this is my folder structure,



Since the beginning I use github as my version control platform and here are some insights,





5.4 Admin Report Implementation

For admins report function I used the mPDF library which was easy to code and it generates pdfs using data directly from the database.



Figure 37 mPDF

Chapter 6: Testing

These are some test cases I used to test and confirm the functionalities of my system.

TEST ID	TEST CASE	INPUT	EXPECTED RESULT	ACTUAL RESULT
TC-01	**Tourist registration with valid data**	Valid email, username, password	Account created successfully	**Pass**
TC-02	**Tourist registration with duplicate email**	Existing email address	Error: account already exists	**Pass**
TC-03	**Login with correct credentials**	Valid username/password	Redirected to role-specific dashboard	**Pass**
TC-04	**Login with incorrect password**	Valid username, wrong password	Error: wrong password	**Pass**
TC-05	**Partner registration pending approval**	New tour guide registration	Account created with `is_approved = 0`, login blocked until approved	**Pass**
TC-06	**Add a new shop listing**	Valid listing form data	Listing created with status "pending"	**Pass**
TC-07	**Edit listing — readonly fields protected**	Attempt to submit form with unchanged readonly fields	Title/location/city unchanged, only description/price/hours updated	**Pass**
TC-08	**Delete a listing**	Click delete on an owned listing	Listing hidden from all views (`is_deleted = 1`), underlying data preserved	**Pass**
TC-09	**Tour guide booking price calculation**	3-day booking, price Rs. 11,000/day	Total price = Rs. 33,000	**Pass**
TC-10	**Hotel booking with multiple rooms**	2 rooms selected, 3 nights	Total price = sum of (room price × nights) for both rooms	**Pass**
TC-11	**Booking with unavailable room**	Room marked unavailable selected	Error: room unavailable	**Pass**
TC-12	**Confirm booking (partner)**	Partner confirms a pending booking	Status updated to "confirmed", notification sent to tourist	**Pass**
TC-13	**Attempt to edit another user's listing**	Listing ID belonging to a different user	Error: unauthorized	**Pass**
TC-14	**Send message to partner**	Tourist sends message from listing page	Message saved, visible in both users' message threads	**Pass**
TC-15	**Write review after completed booking**	Tourist submits review for completed booking	Review saved and displayed on listing page	**Pass**

Figure 38 Testcases Table

Chapter 7: Conclusion

7.1 Achievements

As a conclusion I would like to describe what I achieved from my project, CeylonTourMateLK.

- A fully functional centralized platform connecting all tourists and local service providers
- Role based login to make the system easy to use for users
- A complete way for tourists to go form searching booking confirmation to payments.
- System sends automatic notifications to users(ex: when local partner approves a booking tourist receives a notification)
- Throughout the project I have learned all the knowledge of SDLC process specially with the diagrams and coding part.

7.2 Limitations

- The main limitations is the payment system still implement a payment processing rather than a real payment gateway.
- I need to find users for this platform by advertising.

7.3 Future Developments

- My main goal is to find users and give them the knowledge to how to use this platform and advantages of this one compared to the existing third party sites.
- In the future I would like to integrate this with a real payment gateway for live transactions.
- I would like to build a dedicated mobile app for tourist to connect with this website too.

7.4 Final Conclusion

CeylonTourMateLK will be a great platform for community as well as my self too. From the beginning I have learned everything again to make this project a success, all the things is a original work done my me, my self.

This platform address a huge gap in the tourism industry in srilanka and as a solution for that provide a centralized trustworthy platform for the tourists.

References

- Stack Overflow. (n.d.). *How to model messages exchanged between users? - ER Diagram*. [online] Available at: https://stackoverflow.com/questions/35310283/how-to-model-messages-exchanged-between-users-er-diagram [Accessed 26 July 2026].
- asdad. (2026). *Reddit - please wait for verification*. Reddit.Com. https://www.reddit.com/r/sql/comments/13ppmpj/what_do_you_think_of_this_er_diagram_i_feel_like/?utm_source=chatgpt.com
- table, I. (2014, November 15). *Stack overflow*. Stack Overflow. https://stackoverflow.com/questions/26950903/is-it-okay-for-the-foreign-key-field-to-have-a-different-name-than-the-field-it